

# PIsignHD: A New Structure for the SQIsign Family with Flexible Applicability

Kaizhan Lin<sup>1</sup>[0000–0001–8577–2509], Weize Wang<sup>1</sup>[0009–0000–0516–1776],  
Chang-An Zhao<sup>2,3</sup>[0000–0002–1792–0589], and Yunlei  
Zhao✉<sup>1,4</sup>[0000–0002–2623–9170]

<sup>1</sup> College of Computer Science and Artificial Intelligence, Fudan University,  
Shanghai, China

linkzh@fudan.edu.cn  
wzwang23@m.fudan.edu.cn  
ylzhao@fudan.edu.cn

<sup>2</sup> School of Mathematics, Sun Yat-sen University, Guangzhou, China  
zhaochan3@mail.sysu.edu.cn

<sup>3</sup> Guangdong Key Laboratory of Information Security, Guangzhou, China

<sup>4</sup> State Key Laboratory of Cryptology, Beijing, China

**Abstract.** In this paper, we propose a new structure for the SQIsign family: *Pentagon Isogeny-based Signature in High Dimension* (referred to as PIsignHD). The new structure separates the hash of the commitment and that of the message by employing two cryptographic hash functions. This feature is desirable in reality, particularly for applications based on mobile low-power devices or for those deployed interactively over the Internet or in the cloud computing setting.

This structure can be generally applied to all SQIsign variants. In this work, we focus on the instance based on SQIsignHD. Compared with SQIsignHD, PIsignHD has the same signature size (even smaller for some application scenarios). For the NIST-I security level, the signature size of PIsignHD can be reduced to 519 bits, while the SQIsignHD signature takes 870 bits. Additionally, PIsignHD has an efficient online signing process and enjoys much desirable application flexibility. In our experiments, the online signing process of PIsignHD runs in 4 ms.

**Keywords:** Digital signatures · SQIsign · SQIsignHD · Isogeny ·  $\Gamma$ -protocol.

## 1 Introduction

Isogeny-based cryptography is attractive for its compact keys in post-quantum cryptography, but the expensive computational cost limits the practical applications of isogeny-based cryptosystems. Various digital signatures under isogeny assumptions have been proposed in recent years, such as [23,13,4,19]. However, many of these schemes suffer from relatively large signature or public-key sizes.

Conversely, SQIsign [14] and SQIsignHD [12] fully highlight the compactness as isogeny-based signatures. SQIsign has a very efficient verification, but

the signing phase is expensive due to the ideal-to-isogeny translation. Although the ideal-to-isogeny translation has been improved recently [15,25,29], it remains the main efficiency bottleneck in the signing phase. SQIsignHD applies the algorithms derived from SIDH attacks [6,27,33], and offers a remarkably smaller signature size and much faster response since the signer does not need to compute large degree isogenies. Conversely, the verification in SQIsignHD is inefficient as it involves isogeny computations in high dimension.

**Motivation.** Currently, both SQIsign and SQIsignHD are based on  $\Sigma$ -protocols. Therefore, the challenge is derived from the knowledge of the commitment and the message. However, this feature may result in inconvenient deployments or inefficient implementations, particularly for applications based on low-power devices or applications in the cloud computing setting. We present and discuss some motivating application scenarios below.

- **Application 1: Hardware wallet based on SIM card.** This is a typical application scenario based on mobile low-power devices. The SIM card acts as the signer who keeps the signing secret key and performs signing operations, while the message data to be signed is usually generated by applications in the mobile phone. When generating a signature based on a  $\Sigma$ -protocol, the SIM card has to compute the hash value of the concatenation of the commitment and the message data (note that when the message data is large, this would be unfriendly as the interaction cost is expensive), or transfer the commitment to the system on chip (SoC) to compute the hash value.
- **Application 2: Document online signing by enterprise.** When using the signature scheme in practice, particularly by enterprises, the signing server is usually deployed in the cloud or run by the enterprise. In the scenario of online signing,  $\Sigma$ -signatures require the signer to upload the entire document to the signature server. This may consume a significant amount of bandwidth and cause more computing burden on the signature server, resulting in a system bottleneck.

*Remark 1.* When sending the full message is problematic, one may consider signing a hash of the message instead of the full message. For signing the message  $m$  with  $\Sigma$ -signature, we need to consider the collision resistance of single-hashing  $h(a||m)$ , where  $a$  is the commitment and  $h$  is a hash function. But for signing  $h(m)$ , we need to consider the collision resistance of double-hashing  $h(a||h(m))$ . Double-hashing has much lower collision-resistance than single-hashing. That is the reason why  $\Sigma$ -signatures do not recommend to sign  $h(m)$ .

In 1989, Even, Goldreich and Micali [20] introduced online/offline signatures, which are desirable for low-power devices. The main idea of online/offline signatures is to divide the signature into the online phase and the offline phase. Generally, the online phase is required to be fast as possible, while the offline phase can be connected to the power. In 2013, Yao and Zhao [38] proposed  $\Gamma$ -protocols and a novel transformation method, known as  $\Gamma$ -transformation. Unlike  $\Sigma$ -signatures, the signatures via  $\Gamma$ -transformation separate the hash of the commitment  $a$  and that of the message  $m$ , by employing two secure hash

functions  $h_1$  and  $h_2$  to compute the hash values  $h_1(a)$  and  $h_2(m)$ , respectively. From the target one-way property of  $h_1$ , the value  $h_1(a)$  (or a set of values  $\{h_1(a_1), h_1(a_2), \dots, h_1(a_s)\}$  with commitments  $a_1, a_2, \dots, a_s$ ) can be public or stored on the verifier's side. Consequently, the verifier can precompute some intermediate values that are relevant to the hash values of the commitment to enhance the verification performance. Moreover,  $\Gamma$ -signatures allow the verifier to compute  $h_2(m)$  in advance without the knowledge of the commitment  $a$ . When a trusted verifier would like to request the signer to sign a message  $m$ , it can transfer the hash value  $h_2(m)$  instead of the whole message to the signer, thereby significantly reducing the communication cost and the computational cost of hashing for the signer in the response phase. The specific construction of  $\Gamma$ -signatures also benefits the online response of the signer, since all the intermediate values irrelevant to the message and used to generate the response can be computed offline. As a result,  $\Gamma$ -signatures offer an efficient online structure and enjoy the advantage of application flexibility.

**Contribution.** In this paper, we propose a new structure for the SQIsign family, which is illustrated in Figure 1. The new structure is constructed via  $\Gamma$ -transformation. The main difference between the SQIsign family and our new structure is that the latter contains an additional isogeny  $\varphi_{com} : E_1 \rightarrow E_2$ , which is derived from the knowledge of the commitment. Besides, the challenge isogeny  $\varphi_{chl} : E_A \rightarrow E_3$  is hashed from the knowledge of the message. Correspondingly, the response isogeny is from  $E_2$  to  $E_3$ .

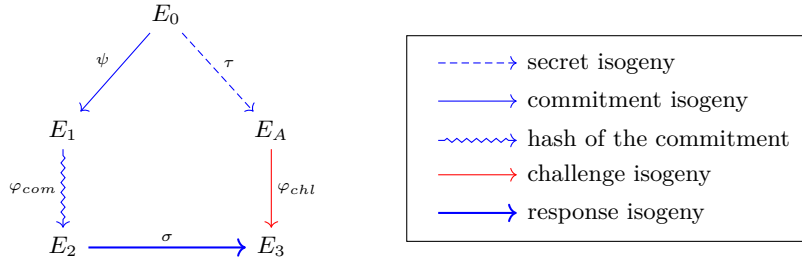


Fig. 1: A sketch of our new structure

Obviously, our new structure can be easily applied to the SQIsign family. To show the advantages of the new structure compared to the traditional structure, we take SQIsignHD as an instance and introduce **Pentagon Isogeny-based Signature in High Dimension** (referred to as PIsignHD or  $\Pi$ -signHD). At first glance, the efficiency of PIsignHD appears to be slightly inferior to SQIsignHD due to the additional isogeny involved. But in normal cases, SQIsignHD and PIsignHD have the same signature size. Furthermore, PIsignHD has the following additional advantages, which are attractive in applications.

- **Flexible challenge generation:** In SQIsignHD, the challenge isogeny is derived from the knowledge of the public key, the commitment and the message. Benefiting from  $\Gamma$ -transformation, the generation of the challenge isogeny in PIsignHD only requires the public key and the message. This feature tackles the applications as we mentioned above. In Applications 1 and 2,

the signature requester can directly transmit the hash value of the message, which reduces transmission and computational requirements for the signer.

- **More compact signature in applications:** The signature of SQIsignHD involves the  $j$ -invariant of a supersingular curve. If the public storage is available, PIsignHD avoids storing it, and the signature size can be reduced from  $6.5\lambda$  bits to around  $3.5\lambda$  bits, where  $\lambda$  is the security parameter.
- **Fast online signing computations:** As previously mentioned, the verifier can transfer the kernel of the challenge isogeny instead of the whole message, saving the time for the signer to hash the message. Besides, the signer can precompute intermediate values that are irrelevant to the message. In our implementation, the online signature computations take only 4 ms.
- **Storage saving:** To adapt the online/offline technique in SQIsignHD, the signer has to store all the intermediate values that are used to sign the message. Conversely, PIsignHD allows some of the values to be public, or stored on the verifier’s size. Therefore, PIsignHD reduces the storage requirements for the signer, which is preferred in applications.

**Related Work.** Recently, Renan and Kutas proposed a quantum-resistant adaptor signature called SQIAsignHD [32]. This scheme underlies SQIsignHD and utilizes the artificial orientation on SIDH [2]. We believe some techniques in this work could also be beneficial for SQIAsignHD with further research. Very Recently, several variants of SQIsignHD are proposed [1,28,17]. Our structure can also be applied to these schemes. More technical details are left as future work.

**Organization.** The remainder of our paper is organized as follows. Section 2 reviews the necessary preliminaries. In Section 3 we propose a high-level overview of PIsignHD and the underlying identification protocol. The security proofs are provided in Section 4. Section 5 introduces the concrete implementation of PIsignHD, and presents the experimental results. Finally we conclude in Section 6.

## 2 Preliminaries

In this section, we recall  $\Gamma$ -protocols and SQIsignHD. The necessary mathematical backgrounds are left in Appendix A, while the introductions of  $\Sigma$ -protocol and Fiat–Shamir paradigm are left in Appendix B.

### 2.1 $\Gamma$ -Protocol and $\Gamma$ -Transformation

Assume that  $\mathcal{P}$  and  $\mathcal{V}$  are probabilistic polynomial time machines, and the advantage of  $\mathcal{P}$  over  $\mathcal{V}$  is that  $\mathcal{P}$  knows  $w$  with  $(x, w) \in \mathcal{R}$ , where  $\mathcal{R}$  is an  $\mathcal{NP}$ -relation. Then  $\Gamma$ -protocol proceeds as follows:

- $\mathcal{P}$  sends a commitment  $a$  and a random string  $d$  to  $\mathcal{V}$ ;
- $\mathcal{V}$  sends a random string  $e$  to  $\mathcal{P}$ ;
- $\mathcal{P}$  sends a reply  $z$  with respect to  $e$ , and  $\mathcal{V}$  accepts or rejects based on  $(x, a, d, e, z)$ .

**Definition 1** ([38]).  $\Gamma$ -protocol is a three-round public-coin protocol  $\langle \mathcal{P}, \mathcal{V} \rangle$  for an  $\mathcal{NP}$ -relation  $\mathcal{R}$  that proceeds as above. Besides,  $\Gamma$ -protocols should satisfy the following properties:

- **Completeness:**  $\mathcal{V}$  always accepts if  $\mathcal{P}$  and  $\mathcal{V}$  follow the protocol.
- **Knowledge extraction:** Given two valid conversations  $(a, d, e, z)$  and  $(a, d', e', z')$  on any input  $x$  with  $(d, e) \neq (d', e')$ , one can recover the witness  $w$  such that  $(x, w) \in \mathcal{R}$  in polynomial time with respect to an  $\mathcal{NP}$ -relation  $R_e$ , referred to as  $e$ -condition, that  $R_e(d, d', e, e', z, z') = 1$ . In particular, setting  $d = d'$  implies that the protocol has the special soundness property<sup>1</sup>.
- **Special honest verifier zero-knowledge (SHVZK):** There exists a probabilistic polynomial-time simulator  $\mathcal{S}$ , which takes as input  $x$  and outputs an accepting conversation  $(a', d', e', z')$ , with the same (or computationally indistinguishable) probability distribution as the conversation  $(a, d, e, z)$  of the real protocol.

$\Gamma$ -transformation can convey a  $\Gamma$ -protocol into a signature scheme. Different from Fiat-Shamir paradigm,  $\Gamma$ -transformation adapts two hash functions  $h_1, h_2$  to compute  $d = h_1(a)$  and  $e = h_2(m)$ , respectively. The verifier accepts if  $d = h_1(a)$  and  $(a, d, e, z)$  is a valid conversation. To be precise,  $\Gamma$ -signatures are demonstrated as follows:

- **Key Generation:** The signer generates  $x = F(w)$  such that  $(x, w) \in \mathcal{R}$  where  $F$  is a one-way and polynomial-time computable function. The public key is  $x$  and the secret key is  $w$ .
- **Signature:** The signer randomly selects  $r_P \in R_P$  and computes  $a = f_a(r_P, x)$ , where  $f_a$  is a polynomial-time computable function. Then, compute  $d = h_1(a)$  where  $h_1$  is a secure hash function. Given a message  $m$ , the signer computes  $e = h_2(m)$ , where  $h_2$  is a secure hash function. From  $(w, a, d, e)$  the signer generates  $z$ , and finally outputs  $(a, d, z)$  as the signature<sup>2</sup>.
- **Verification:** Given  $m$ , the verifier computes  $e = h_2(m)$ . The verifier accepts if  $d = h_1(a)$  and  $(a, d, e, z)$  is a valid conversation, according to the polynomial-time computable verification procedure for the underlying  $\Gamma$ -protocol.

## 2.2 SQIsignHD

SQIsignHD is a compact and post-quantum signature scheme introduced by Dartois, Leroux, Rebert and Wesolowski [11]. It is constructed from an identification protocol via Fiat-Shamir paradigm. Currently, there are two versions of SQIsignHD: FastSQIsignHD and RigorousSQIsignHD. We only focus on constructing a fast online signature scheme based on FastSQIsignHD for efficiency. The identification protocol underlying FastSQIsignHD proceeds as follows:

<sup>1</sup> The definition here is slightly different from that of [38]. They limit that the knowledge extracts when  $\mathcal{R}_e(d, d', e, e') = 1$ , where  $\mathcal{R}_e$  is an  $\mathcal{NP}$ -condition. However,  $\Gamma$ -protocol only requires that the  $e$ -condition holds with overwhelming probability.

<sup>2</sup> In some specific signature schemes, such as  $\Gamma$ -signatures for DLP [38], the signature can be compressed by  $(d, z)$  since  $a$  can be computed according to  $(d, z)$ .

- **Setup:** Select a prime  $p = c \cdot \ell^f \cdot \ell'^{f'} - 1$ , where  $c$  is a small cofactor and  $\ell^f \approx \ell'^{f'} \approx 2^\lambda$  with  $\lambda$  the security level. Define a supersingular elliptic curve  $E_0$  whose endomorphism ring is known. Let  $g$  be an integer big enough but smaller than  $f$ .
- **Keygen:** The prover generates a random isogeny walk  $\tau : E_0 \rightarrow E_A$  of degree  $\ell'^\bullet \approx p$  and an equivalent isogeny  $\tau' : E_0 \rightarrow E_A$  of degree  $\ell^\bullet \approx p$ . The public key is the elliptic curve  $E_A$  and the secret key is  $(\tau, \tau')$ .
- **Commitment:** The prover generates a random (secret) isogeny walk  $\psi : E_0 \rightarrow E_1$  of degree  $\ell'^\bullet \approx p$ . Afterwards, the prover sends  $E_1$  to the verifier.
- **Challenge:** The verifier generates a random isogeny walk  $\varphi : E_A \rightarrow E_2$  of degree  $\ell'^{f'}$  and sends the description of  $\varphi$  to the prover.
- **Response:** From the knowledge of the secret key, the commitment and the challenge, the prover generates a new isogeny  $\sigma : E_1 \rightarrow E_2$  of degree  $q$  such that  $q$  is  $\ell^g$ -good, i.e.,  $\ell^g - q$  is a prime congruent to 1 modulo 4. Then the prover computes  $\sigma(P_1)$  and  $\sigma(Q_1)$  where  $\{P_1, Q_1\}$  is the canonical basis of  $E_1[\ell^f]$ , and sends  $(q, \sigma(P_1), \sigma(Q_1))$  to the verifier.
- **Verify:** the verifier generates the canonical basis  $\{P_1, Q_1\}$  of  $E_1[\ell^f]$ . Then the verifier accepts if  $(E_1, E_2, q, (P_1, Q_1), (\sigma(P_1), \sigma(Q_1)))$  correctly represents a  $q$ -isogeny  $\sigma$  from  $E_1$  to  $E_2$ .

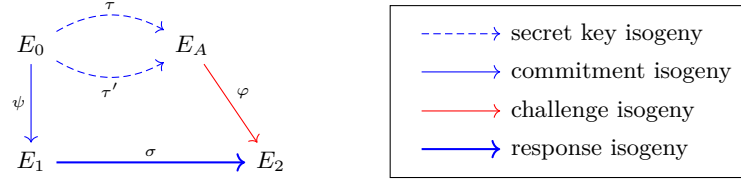


Fig. 2: A sketch of the SQIsignHD identification protocol

Compared with SQIsign, SQIsignHD avoids the complex ideal-to-isogeny translation. The main procedures are illustrated in Algorithm 1.

---

**Algorithm 1** FastRespond [11, Algorithm 2]

---

**Require:** The isogenies  $\tau, \tau' : E_0 \rightarrow E_A$  of degree  $\ell'^\bullet$  and  $\ell^\bullet$  respectively, the ideals  $I_\tau$  and  $I_{\tau'}$  associated to  $\tau$  and  $\tau'$  respectively, the isogeny  $\psi : E_0 \rightarrow E_1$  of degree  $\ell'^\bullet$ , the ideal  $I_\psi$  associated to  $\psi$ , the isogeny  $\varphi : E_A \rightarrow E_2$  of degree  $\ell'^{f'}$ .

**Ensure:**  $(\sigma(P_1), \sigma(Q_1), q)$  where  $(P_1, Q_1)$  is the canonically determined basis of  $E_1[\ell^f]$  and  $\sigma : E_1 \rightarrow E_2$  is an isogeny of  $\ell^g$ -good degree  $q$  coprime to  $\ell$ .

- 1:  $I_\varphi \leftarrow \mathbf{IsogenyToIdeal}(\ker(\varphi), \tau', I_{\tau'}), J \leftarrow \overline{I_\psi} \cdot I_\tau \cdot I_\varphi$ ;
  - 2:  $I \leftarrow \mathbf{RandomEquivalentIdeal}_{\ell^g}(J)$  and compute the reduced norm  $q$  of  $I$ ;
  - 3: If  $q$  is not  $\ell^g$ -good or  $\gcd(q, \ell') \neq 1$ , go back to Line 2;
  - 4: Compute the canonical basis of  $\{P_1, Q_1\}$  of  $E_1[\ell^f]$ ;
  - 5:  $(\sigma(P_1), \sigma(Q_1)) \leftarrow \mathbf{EvalTorsion}_{\ell^f}(I, P_1, Q_1, \psi, \varphi \circ \tau, I_\psi, I_\tau \cdot I_\varphi)$ ;
  - 6: **return**  $(\sigma(P_1), \sigma(Q_1), q)$ .
- 

The following are the sub-algorithms applied in Algorithm 1:

- **IsogenyToIdeal**( $\ker(\varphi), \tau', I_{\tau'}$ ): Given the kernel of an isogeny  $\varphi : E_A \rightarrow E_2$ , an isogeny  $\tau' : E_0 \rightarrow E_A$  of degree coprime to  $\deg(\varphi)$  and the corresponding ideal  $I_{\tau'} \subset \mathcal{O}$ , outputs the ideal  $I_\varphi$  associated to  $\varphi$ ;

- **RandomEquivalentIdeal** $_{\ell^g}(J)$ : Given an ideal  $J$ , outputs an equivalent ideal  $I$  that is uniformly random among ideals of norm  $\leq \ell^g$ ;
- **EvalTorsion** $_{\ell^f}(I, P_1, Q_1, \rho_1, \rho_2, I_{\rho_1}, I_{\rho_2})$ : Given an ideal  $I$ , a basis  $\{P_1, Q_1\}$  of  $E_1[\ell^f]$ , and two isogenies  $\rho_1: E_0 \rightarrow E_1$  and  $\rho_2: E_0 \rightarrow E_2$  and the corresponding ideals  $I_{\rho_1}, I_{\rho_2}$ , outputs  $\sigma(P_1)$  and  $\sigma(Q_1)$ , where  $\sigma$  is the isogeny associated to  $I$ .

In the response phase, the prover should evaluate the isogeny  $\sigma$  on the basis  $\{P_1, Q_1\}$ . Since the degree of  $\sigma$  is non-smooth in general, it is difficult to evaluate the isogeny directly with Vélú's formula [36,3]. However, note that the prover has the knowledge of the smooth degree isogenies from  $E_0$  to  $E_1$  and  $E_2$ , respectively, i.e.,  $\psi: E_0 \rightarrow E_1$  and  $\varphi \circ \tau: E_0 \rightarrow E_2$ . Furthermore, the endomorphism ring of  $E_0$  is known. Assuming  $\mathcal{O}\gamma = I_\psi \cdot I_\sigma \cdot \overline{I_\tau} \cdot \overline{I_\varphi}$ , it is easy to prove that

$$\sigma = \frac{\varphi \circ \tau \circ \gamma \circ \hat{\psi}}{[\deg(\varphi) \deg(\tau) \deg(\psi)]}. \quad (1)$$

Therefore, the prover can evaluate  $\sigma(P_1)$  and  $\sigma(Q_1)$  efficiently.

At first glance, the prover still has to evaluate several isogenies to generate the response. Fortunately, the current implementation of SQIsignHD applies a more elegant approach to eliminate almost all the isogeny computations. We provide a detailed review of the current implementation in Appendix C.

### 3 PIsignHD

In this section we propose the PIsignHD identification protocol, and the PIsignHD digital signature via  $\Gamma$ -transformation.

#### 3.1 Identification protocol

Let  $\lambda$  be a security parameter. The PIsignHD identification protocol goes as follows:

- **Setup**: Select a prime  $p = c \cdot \ell^f \cdot \ell'^{f'} - 1$ , where  $c$  is a small cofactor and  $\ell^f \approx \ell'^{f'} \approx 2^\lambda$  with  $\lambda$  the security level. Define a supersingular elliptic curve  $E_0$  whose endomorphism ring is known. Let  $g$  be an integer big enough but smaller than  $f$ .
- **Keygen**: The prover generates a random isogeny walk  $\tau: E_0 \rightarrow E_A$  of degree  $\ell'^\bullet \approx p$  and an equivalent isogeny  $\tau': E_0 \rightarrow E_A$  of degree  $\ell^\bullet \approx p$ . The public key is the elliptic curve  $E_A$  and the secret key is  $(\tau, \tau')$ .
- **Commitment**: The prover generates a random isogeny walk  $\psi: E_0 \rightarrow E_1$  of degree  $\ell'^\bullet \approx p$  and an equivalent isogeny  $\psi': E_0 \rightarrow E_1$  of degree  $\ell^\bullet \approx p$ , and then selects a random cyclic isogeny walk  $\varphi_{com}: E_1 \rightarrow E_2$  of degree  $\ell'^{f'}$ . Afterwards, the prover sends  $E_1$  and the description of  $\varphi_{com}$  to the verifier.
- **Challenge**: The verifier generates a random isogeny walk  $\varphi_{chl}: E_A \rightarrow E_3$  of degree  $\ell'^{f'}$  and sends the description of  $\varphi_{chl}$  to the prover.

- **Response:** From the knowledge of the secret key, the commitment and the challenge, the prover generates a new isogeny  $\sigma : E_2 \rightarrow E_3$  of degree  $q$  such that  $q$  is  $\ell^g$ -good and coprime to  $\ell'$ , and computes  $\sigma(P_2)$  and  $\sigma(Q_2)$  where  $\{P_2, Q_2\}$  is the canonical basis of  $E_2[\ell^f]$ . Then the prover sends  $R = (q, \sigma(P_2), \sigma(Q_2))$  to the verifier.
- **Verify:** the verifier generates the canonical basis  $\{P_2, Q_2\}$  of  $E_2[\ell^f]$ . Then the verifier accepts if  $(E_2, E_3, q, (P_2, Q_2), (\sigma(P_2), \sigma(Q_2)))$  correctly represents a  $q$ -isogeny  $\sigma$  from  $E_2$  to  $E_3$ .

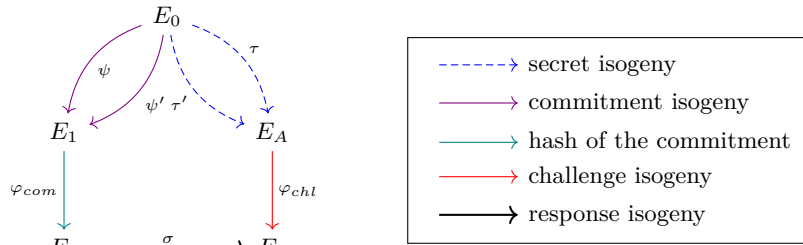


Fig. 3: A sketch of PSignHD

The completeness property of our  $\Gamma$ -protocol is obvious. The security proofs of the knowledge extraction property and the zero-knowledge property are left in Section 4.

### 3.2 Digital signature

Via  $\Gamma$ -transformation, PSignHD is derived by the identification protocol in Section 3.1. The setup and the key generation phases are identical to those of the identification protocol. The signature and the verification proceed as follows:

- **Sign:**  $(sk, m) \rightarrow \Sigma$  Pick a random (secret) isogeny  $\psi : E_0 \rightarrow E_1$  of degree  $\ell'^\bullet \approx p$  and an equivalent isogeny  $\psi' : E_0 \rightarrow E_1$  of degree  $\ell'^\bullet \approx p$ . Then, construct the cyclic isogeny  $\varphi_{com} : E_1 \rightarrow E_2$  with respect to the hash of  $E_1$ . From the hash of  $m$ , construct the isogeny  $\varphi_{chl} : E_A \rightarrow E_3$ . Finally, generate a new isogeny  $\sigma : E_2 \rightarrow E_3$  and compute the corresponding pairs  $R = (\sigma(P_2), \sigma(Q_2), q)$  with  $\{P_2, Q_2\}$  the canonical basis of  $E_2[\ell^f]$  and  $q$  coprime to  $\ell'$ . The signature is  $(E_1, R)$ .
- **Verify:**  $(pk, m, \Sigma) \rightarrow \text{True or False}$  Parse  $\Sigma$  as  $(E_1, R)$ , where  $R = (\sigma(P_2), \sigma(Q_2), q)$ . Firstly, compute the isogeny  $\varphi_{com} : E_1 \rightarrow E_2$  which is hashed from the knowledge of  $E_1$ . From the message  $m$ , construct the isogeny  $\varphi_{chl} : E_A \rightarrow E_3$ . Generate the determined canonical basis  $\{P_2, Q_2\}$  of  $E_2[\ell^f]$ , and accept if  $(E_2, E_3, q, (P_2, Q_2), (\sigma(P_2), \sigma(Q_2)))$  correctly represents a  $q$ -isogeny  $\sigma : E_2 \rightarrow E_3$ .



In the signing and verifying procedures, the isogenies  $\varphi_{com}$  and  $\varphi_{chl}$  are generated by hashing. To achieve this, we first define a secure hash function  $\mathcal{H} : \{0, 1\}^* \rightarrow [1, \mu]$ , where  $\mu = \ell'^{f'-1}(\ell' + 1)$ . Same as SQIsign and SQIsignHD, we use the secure hash function  $\mathcal{H}'$  defined in [16, Section 3.1], which is derived from [8]. Taking a supersingular curve  $E$  and an integer as inputs, the hash function  $\mathcal{H}'$  outputs a cyclic  $\ell'^{f'}$ -isogeny with domain  $E$ . In practice, we set  $\varphi_{com} = \mathcal{H}'(E_1, \mathcal{H}(j(E_1)))$  and  $\varphi_{chl} = \mathcal{H}'(E_A, \mathcal{H}(m))$ .

## 4 Security Proof

In this section we present the security proofs of PIsignHD. The proof of the completeness property is omitted as it is obvious. In the following, we focus on the proofs of the knowledge extraction property and the zero-knowledge property. The knowledge extraction proof is similar to the special soundness proof of the SQIsignHD identification protocol, but we need to prove that the  $e$ -condition holds with overwhelming probability as well. Several lemmas will be proposed to adequately illustrate this issue. The zero-knowledge proof parallels that of the SQIsignHD identification protocol, particularly we use the same oracle (Definition 2) to construct the simulator.

### 4.1 Knowledge extraction

Recall the knowledge extraction property of  $\Gamma$ -protocols: Given two pairs of valid conversations  $(a, d, e, z)$  and  $(a, d', e', z')$  on any input  $x$  with  $(d, e) \neq (d', e')$ , one can recover the witness  $w$  such that  $(x, w) \in \mathcal{R}$  in polynomial time with respect to an  $\mathcal{NP}$ -relation  $\mathcal{R}_e$ , referred to as the  $e$ -condition, that  $\mathcal{R}_e(d, d', e, e', z, z') = 1$ .

In the PIsignHD identification protocol, the commitment is  $E_1$ , while  $\varphi_{com}$  is a random isogeny starting from  $E_1$ . The challenge corresponds to  $\varphi_{chl} : E_A \rightarrow E_3$ , and the response is of form  $R = (q, \sigma(P_2), \sigma(Q_2))$ , where  $q$  is the degree of the response isogeny  $\sigma : E_2 \rightarrow E_3$  and  $(\sigma(P_2), \sigma(Q_2))$  are the images of the torsion basis  $\{P_2, Q_2\}$  of  $E_2[\ell^f]$  by  $\sigma$ . The hard problem underlying the knowledge extraction property is known as *Supersingular Endomorphism Problem*:

*Problem 1 (Supersingular Endomorphism Problem).* Given a prime  $p$  and a supersingular elliptic curve  $E$  over  $\mathbb{F}_{p^2}$ , find a non-trivial endomorphism of  $E$  that can be efficiently evaluated.

When  $\varphi_{com_1}$  and  $\varphi_{com_2}$  in the two pairs of valid conversations  $(E_1, \varphi_{com_1}, \varphi_{chl_1}, R_1)$  and  $(E_1, \varphi_{com_2}, \varphi_{chl_2}, R_2)$  are equivalent, the knowledge extraction property is reduced to the special soundness property. In this situation, the proof is almost consistent with the special soundness proofs of the SQIsignHD identification protocol. Similarly, it is easy to prove the knowledge extraction property when the challenges of the valid conversations are equivalent. When  $\varphi_{com_1} \neq \varphi_{com_2}$  and  $\varphi_{chl_1} \neq \varphi_{chl_2}$ , one can also extract the knowledge under the  $e$ -condition that:  $\mathcal{R}_e(\varphi_{com_1}, \varphi_{com_2}, \varphi_{chl_1}, \varphi_{chl_2}, \sigma_1, \sigma_2) = 1$  iff there does not exist  $s \in \mathbb{Z}$  such that  $[s] = \hat{\varphi}_{chl_2} \circ \sigma_2 \circ \varphi_{com_2} \circ \hat{\varphi}_{com_1} \circ \hat{\sigma}_1 \circ \varphi_{chl_1}$ .

**Proposition 1.** Let  $(E_1, \varphi_{com_1}, \varphi_{chl_1}, R_1)$  and  $(E_1, \varphi_{com_2}, \varphi_{chl_2}, R_2)$  be two pairs of accepting conversations, where  $R_1 = (q_1, \sigma_1(P_2), \sigma_1(Q_2))$  and  $R_2 = (q_2, \sigma_2(P'_2), \sigma_2(Q'_2))$  with  $\langle P_2, Q_2 \rangle = E_2[\ell^f]$  and  $\langle P'_2, Q'_2 \rangle = E'_2[\ell^f]$ . If  $(\varphi_{com_1}, \varphi_{chl_1}) \neq (\varphi_{com_2}, \varphi_{chl_2})$ , then one can compute a non-trivial endomorphism of  $E_A$  that can be efficiently evaluated with respect to the  $e$ -condition that:  $\mathcal{R}_e(\varphi_{com_1}, \varphi_{com_2}, \varphi_{chl_1}, \varphi_{chl_2}, \sigma_1, \sigma_2) = 1$  iff there does not exist  $s \in \mathbb{Z}$  such that  $[s] = \hat{\varphi}_{chl_2} \circ \sigma_2 \circ \varphi_{com_2} \circ \hat{\varphi}_{com_1} \circ \hat{\sigma}_1 \circ \varphi_{chl_1}$ . If  $\varphi_{com_1} = \varphi_{com_2}$  or  $\varphi_{chl_1} = \varphi_{chl_2}$ , then the  $e$ -condition always holds. Especially, the PSignHD identification protocol has the special soundness property.

*Proof.* Since the two conversations are valid, one can obtain the knowledge of the response isogenies  $\sigma_1 : E_2 \rightarrow E_3$  and  $\sigma_2 : E'_2 \rightarrow E'_3$ . Note that  $\varphi_{com_1} : E_1 \rightarrow E_2$ ,  $\varphi_{com_2} : E_1 \rightarrow E'_2$ ,  $\varphi_{chl_1} : E_A \rightarrow E_3$  and  $\varphi_{chl_2} : E_A \rightarrow E'_3$  are known. As illustrated in Figure 4,  $\alpha = \hat{\varphi}_{chl_2} \circ \sigma_2 \circ \varphi_{com_2} \circ \hat{\varphi}_{com_1} \circ \hat{\sigma}_1 \circ \varphi_{chl_1}$  is an endomorphism of  $E_A$  that can be efficiently evaluated.

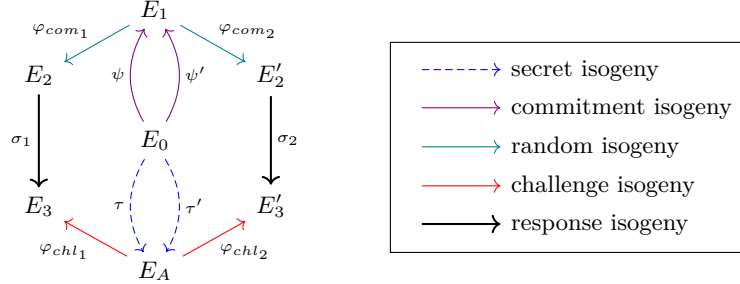


Fig. 4: Knowledge extraction

If the  $e$ -condition holds, then the endomorphism  $\alpha = \hat{\varphi}_{chl_2} \circ \sigma_2 \circ \varphi_{com_2} \circ \hat{\varphi}_{com_1} \circ \hat{\sigma}_1 \circ \varphi_{chl_1}$  is non-trivial. Now we prove that the  $e$ -condition always holds if  $\varphi_{com_1} = \varphi_{com_2}$  or  $\varphi_{chl_1} = \varphi_{chl_2}$ .

We first prove the endomorphism  $\alpha$  is non-trivial if  $\varphi_{com_1} = \varphi_{com_2}$ . In this case  $\alpha = [\ell'^{f'}] \hat{\varphi}_{chl_2} \circ \sigma_2 \circ \hat{\sigma}_1 \circ \varphi_{chl_1}$ . Suppose for contradiction that  $\alpha' = \hat{\varphi}_{chl_2} \circ \sigma_2 \circ \hat{\sigma}_1 \circ \varphi_{chl_1} = [s]$  with  $s \in \mathbb{Z}$ . Therefore, we have  $q_1 q_2 \ell'^{2f'} = s^2$ . Then

$$[\ell'^{f'} q_2] \circ \hat{\sigma}_1 \circ \varphi_{chl_1} = \hat{\sigma}_2 \circ \varphi_{chl_2} \circ \alpha' = [s] \circ \hat{\sigma}_2 \circ \varphi_{chl_2}.$$

Let  $s = \ell'^{f'} \cdot s'$  with  $s'$  coprime to  $\ell'$ . Then we have  $[q_2] \circ \hat{\sigma}_1 \circ \varphi_{chl_1} = [s'] \circ \hat{\sigma}_2 \circ \varphi_{chl_2}$ . Since  $q_1, q_2$  and  $s'$  are coprime to  $\ell'$ , it follows that  $\ker(\varphi_{chl_1}) = \ker(\varphi_{chl_2})$ . This contradicts the fact that  $(\varphi_{com_1}, \varphi_{chl_1}) \neq (\varphi_{com_2}, \varphi_{chl_2})$  and  $\varphi_{com_1} = \varphi_{com_2}$ . Therefore, the  $e$ -condition holds and the PSignHD identification protocols has the special soundness property.

Assume that  $\varphi_{chl_1} = \varphi_{chl_2}$ . We would like to prove that  $\alpha$  is also non-trivial. Clearly, the endomorphism  $\beta = \hat{\varphi}_{com_2} \circ \hat{\sigma}_2 \circ \varphi_{chl_2} \circ \hat{\varphi}_{chl_1} \circ \sigma_1 \circ \varphi_{com_1} = [\ell'^{f'}] \hat{\varphi}_{com_2} \circ \hat{\sigma}_2 \circ \sigma_1 \circ \varphi_{com_1}$  of  $E_1$  is trivial iff  $\alpha$  is trivial. Suppose that  $\beta$  is trivial. Similar to the previous proof, one can deduce that  $\ker(\varphi_{com_1}) = \ker(\varphi_{com_2})$ . This

is a contradiction because  $(\varphi_{com_1}, \varphi_{chl_1}) \neq (\varphi_{com_2}, \varphi_{chl_2})$  and  $\varphi_{chl_1} = \varphi_{chl_2}$ . Therefore, when  $\varphi_{chl_1} = \varphi_{chl_2}$  the endomorphism  $\beta$  must be non-trivial, i.e., the endomorphism  $\alpha$  is non-trivial, which completes the proof.  $\square$

It remains to prove the  $e$ -condition holds with overwhelming probability, i.e.,

$$\Pr[\mathcal{R}_e(\varphi_{com_1}, \varphi_{com_2}, \varphi_{chl_1}, \varphi_{chl_2}, \sigma_1, \sigma_2) = 0] \leq \text{negl}(\lambda),$$

where  $\text{negl}(\cdot)$  is a negligible function. This confirms that even if  $\varphi_{com_1} \neq \varphi_{com_2}$  and  $\varphi_{chl_1} \neq \varphi_{chl_2}$  (which is the common scenario in practice), the secret key can be extracted with overwhelming probability once the prover adapts the same commitment. In the following, we present Proposition 2 to tackle this problem. To prove Proposition 2, we first propose Lemmas 1, 2 and 3.

**Lemma 1.** *Let  $\Phi_1 = [\ell'^{t_1}]\Phi'_1$ ,  $\Phi_2 = [\ell'^{t_2}]\Phi'_2$  be two isogenies of degree  $(\ell')^{2f'}$ , where  $\Phi'_1 : E_1 \rightarrow E_2$  and  $\Phi'_2 : E_3 \rightarrow E_4$  are cyclic. Assume that  $\sigma : E_2 \rightarrow E_3$  and  $\sigma' : E_4 \rightarrow E_1$  are a  $q_1$ -isogeny and a  $q_2$ -isogeny with  $\gcd(q_1, \ell') = 1$  and  $\gcd(q_2, \ell') = 1$ , respectively. If  $\sigma' \circ \Phi_2 \circ \sigma \circ \Phi_1$  is a trivial endomorphism of  $E_1$ , i.e., there exists  $s \in \mathbb{Z}$  such that  $[s] = \sigma' \circ \Phi_2 \circ \sigma \circ \Phi_1$ , then*

- $t_1 = t_2$ ;
- $[\sigma]_* \widehat{\Phi'_1} = \Phi'_2$ ,  $[\sigma']_* \widehat{\Phi'_2} = \Phi'_1$ ;

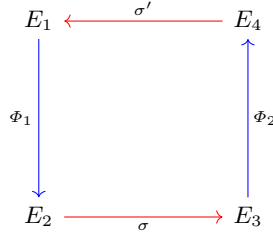


Fig. 5: A sketch of Lemma 1

*Proof.* From  $\Phi_1 = [\ell'^{t_1}]\Phi'_1$ ,  $\Phi_2 = [\ell'^{t_2}]\Phi'_2$ , we have

$$[(\ell')^{2f' - t_1 - t_2} s'] = \sigma' \circ \Phi'_2 \circ \sigma \circ \Phi'_1$$

for some  $s' = \sqrt{q_1 q_2} \in \mathbb{Z}$  which is coprime to  $\ell'$ .

We first prove  $t_1 = t_2$ . Without loss of generality, assume that  $t_1 < t_2$ . Since  $\Phi'_1$  is cyclic, suppose that  $\ker(\Phi'_1) = \langle P \rangle$  where  $P \in E_1[(\ell')^{2f' - 2t_1}]$ . Then, the endomorphism  $[(\ell')^{2f' - t_1 - t_2} s'] = \sigma' \circ \Phi'_2 \circ \sigma \circ \Phi'_1$  sends  $P$  to the point at infinity. It implies that  $2f' - t_1 - t_2 \geq 2f' - 2t_1$ , i.e.,  $t_1 \geq t_2$ , which is a contradiction.

Now we prove the second claim. Suppose that  $Q \in E_1[(\ell')^{2f' - 2t_1}]$  such that  $\langle P, Q \rangle = E_1[(\ell')^{2f' - 2t_1}]$ . Then  $\ker(\widehat{\Phi'_1}) = \langle \Phi'_1(Q) \rangle$ . From  $t_1 = t_2$ , we have  $[(\ell')^{2f' - 2t_1} s'] = \sigma' \circ \Phi'_2 \circ \sigma \circ \Phi'_1$  and thus  $\sigma' \circ \Phi'_2 \circ \sigma \circ \Phi'_1(Q) = \infty_{E_1}$ , i.e.,  $\ker(\widehat{\Phi'_1}) \subset \ker(\sigma' \circ \Phi'_2 \circ \sigma)$ . Since  $\sigma$  and  $\sigma'$  have degrees coprime to  $\ell'$ ,  $\sigma(\ker(\widehat{\Phi'_1})) \subset \ker(\Phi'_2)$ . It follows from  $t_1 = t_2$  that  $|\ker(\Phi'_2)| = |\sigma(\ker(\widehat{\Phi'_1}))|$ . Therefore,  $\ker(\Phi'_2) = \sigma(\ker(\widehat{\Phi'_1}))$ , i.e.,  $[\sigma]_* \widehat{\Phi'_1} = \Phi'_2$ . Analogously, one can imply the other deduction. This ends the proof.  $\square$

*Remark 2.* Lemma 1 shows that if the  $e$ -condition does not hold, then  $\varphi_{chl_2} \circ \hat{\varphi}_{chl_1}$  is the pushforward isogeny of  $\varphi_{com_2} \circ \hat{\varphi}_{com_1}$  through  $\sigma_1$ . Conversely, from  $[\sigma_1]_*(\varphi_{com_2} \circ \hat{\varphi}_{com_1}) = \varphi_{chl_2} \circ \hat{\varphi}_{chl_1}$ , we cannot deduce that the endomorphism  $\alpha = \hat{\varphi}_{chl_2} \circ \sigma_2 \circ \varphi_{com_2} \circ \hat{\varphi}_{com_1} \circ \hat{\sigma}_1 \circ \varphi_{chl_1}$  is trivial. For example, if  $q_1 = \deg(\sigma_1)$  and  $q_2 = \deg(\sigma_2)$  are coprime, then in the proof of Lemma 1 the value  $s' = \sqrt{q_1 q_2} \notin \mathbb{Z}$ . In this case the  $e$ -condition always holds. Therefore, the probability that the  $e$ -condition does not hold is less than that of  $[\sigma_1]_*(\varphi_{com_2} \circ \hat{\varphi}_{com_1}) = \varphi_{chl_2} \circ \hat{\varphi}_{chl_1}$ .

**Lemma 2.** *Let  $\rho_1, \rho_2, \rho_3, \rho_4$  be cyclic  $\ell'^{f'}$ -isogenies chosen uniformly at random,  $\Phi_1 = \rho_2 \circ \rho_1$  and  $\Phi_2 = \rho_4 \circ \rho_3$ . If  $\sigma$  is a  $q$ -isogeny such that  $\gcd(q, \ell') = 1$ , then  $\Pr[[\sigma]_*\Phi_1 = \Phi_2] < (f' + 1)(\ell')^{-2f'}$ .*

*Proof.* Suppose that  $\Phi_1 = \rho_2 \circ \rho_1 = [\ell'^{t_1}]\Phi'_1$  and  $\Phi_2 = \rho_4 \circ \rho_3 = [\ell'^{t_2}]\Phi'_2$  with  $\Phi'_1$  and  $\Phi'_2$  cyclic. To satisfy  $[\sigma]_*\Phi_1 = \Phi_2$ , we have  $t_1 = t_2$  and  $\sigma(\ker(\Phi'_1)) = \ker(\Phi'_2)$ . Since  $\rho_1, \rho_2, \rho_3, \rho_4$  are chosen uniformly at random,

$$\Pr[t_i = u] = \begin{cases} \frac{\ell'}{\ell' + 1}, & \text{if } u = 0, \\ \frac{\ell' - 1}{(\ell' + 1)(\ell')^u}, & \text{if } 0 < u < f', \\ \frac{(\ell')^{1-f'}}{\ell' + 1}, & \text{if } u = f'. \end{cases}$$

where  $i = 1, 2$ . On the other hand, we have

$$\Pr[[\sigma]_*\Phi'_1 = \Phi'_2 | t_1 = t_2 = u] = \begin{cases} \frac{(\ell')^{-2f'+2u+1}}{\ell' + 1}, & \text{if } 0 \leq u < f', \\ 1, & \text{if } u = f'. \end{cases}$$

Therefore, the probability that  $[\sigma]_*\Phi_1 = \Phi_2$  is

$$\begin{aligned} \Pr[[\sigma]_*\Phi_1 = \Phi_2] &= \sum_{u=0}^{f'} \Pr[t_1 = u] \cdot \Pr[t_2 = u] \cdot \Pr[[\sigma]_*\Phi'_1 = \Phi'_2 | t_1 = t_2 = u] \\ &= \sum_{u=0}^{f'} \Pr[t_1 = u]^2 \cdot \Pr[[\sigma]_*\Phi'_1 = \Phi'_2 | t_1 = t_2 = u] \\ &= \frac{(\ell')^{-2f'+3}}{(\ell' + 1)^3} + \sum_{u=1}^{f'-1} \frac{(\ell' - 1)^2 (\ell')^{-2f'+1}}{(\ell' + 1)^3} + \frac{(\ell')^{-2f'+2}}{(\ell' + 1)^2} \\ &< (\ell')^{-2f'} + (f' - 1)(\ell')^{-2f'} + (\ell')^{-2f'} \\ &= (f' + 1)(\ell')^{-2f'}, \end{aligned}$$

which completes the proof.  $\square$

**Lemma 3.** *Let  $P_1$  and  $P_2$  be points of order  $\ell'^{f'}$  defined on  $E_1$  and  $E_2$ , respectively. Assume that  $E_1, E_2$  are supersingular and  $\sigma' : E_1 \rightarrow E_2$  is an isogeny whose degree is coprime to  $\ell'$ . If the endomorphism ring of  $E_2$  is known, then one can generate an endomorphism  $\omega$  of  $E_2$  such that  $\omega \circ \sigma'(P_1) = P_2$  in polynomial time.*

*Proof.* Let  $\text{End}(E_2)$  be the endomorphism ring of  $E_2$ . Suppose that  $\{\theta_1, \theta_2, \theta_3, \theta_4\}$  is a basis of  $\text{End}(E_2)$  that can be evaluated at any point of  $E_2$  in polynomial time. Since  $\text{End}(E_2) \otimes \mathbb{Z}/\ell'^{f'}\mathbb{Z}$  is isomorphic to  $\mathbb{M}_2(\mathbb{Z}/\ell'^{f'}\mathbb{Z})$ , there exist two endomorphisms in the basis  $\{\theta_1, \theta_2, \theta_3, \theta_4\}$ , mapping  $\sigma'(P_1)$  to points that are linearly independent. For simplicity we assume that  $\langle \theta_1(\sigma'(P_1)), \theta_2(\sigma'(P_1)) \rangle = E_2[\ell'^{f'}]$ . This implies the existence of coefficients  $s_1, s_2 \in \mathbb{Z}/\ell'^{f'}\mathbb{Z}$  satisfying that

$$P_2 = [s_1]\theta_1(\sigma'(P_1)) + [s_2]\theta_2(\sigma'(P_1)).$$

Let  $\omega = s_1\theta_1 + s_2\theta_2$ . Then  $\omega \circ \sigma' : E_1 \rightarrow E_2$  is the desired isogeny that sends  $P_1$  to  $P_2$ .  $\square$

**Proposition 2.** *Any prover  $\mathcal{P}(\tau, c)$  (where  $\tau$  is the secret key and  $c$  is the random coin) that can correctly execute the PIsignHD identification protocol and interact with the verifier ensures the knowledge extraction of the protocol.*

*Proof.* From Proposition 1, we only need to prove that the  $e$ -condition holds with overwhelming probability. Suppose that  $\mathcal{P}(\tau, c)$  can break the  $e$ -condition with non-negligible probability for contradiction. Consider Algorithm 2 as follows:

---

**Algorithm 2** Interaction

---

**Require:** The prover  $\mathcal{P}(\tau; c)$  that correctly executes the PIsignHD identification protocol.

**Ensure:** Uniformly randomly selected  $\ell'^{f'}$ -isogenies  $\hat{\phi}_1, \phi_2, \hat{\varphi}_1, \varphi_2$ , and  $q$ -isogenies  $\sigma$  connecting the initial curves of  $\hat{\phi}_1$  and  $\hat{\varphi}_1$  such that  $\gcd(q, \ell') = 1$ .

- 1: Initialize a random coin  $c$  and a verifier  $\mathcal{V}$  that correctly executes the PIsignHD identification protocol, and select a private key  $\tau$ ;
  - 2: Run  $\mathcal{P}(\tau; c)$  in interaction with  $\mathcal{V}$ , and record  $(\psi, \phi_1, \varphi_1, \sigma_1)$ ;
  - 3: Adjust the random coin to obtain  $c'$  such that  $\mathcal{P}(\tau; c')$  uses the same commitment isogeny as  $\mathcal{P}(\tau; c)$ ;
  - 4: Run  $\mathcal{P}(\tau; c')$  in interaction with  $\mathcal{V}$ , and record  $(\psi, \phi_2, \varphi_2, \sigma_2)$ ;
  - 5: **return**  $(\hat{\phi}_1, \phi_2, \hat{\varphi}_1, \varphi_2, \sigma_1, \sigma_2)$ .
- 

From the assumption, the endomorphism  $\alpha = \hat{\varphi}_{chl_2} \circ \sigma_2 \circ \varphi_{com_2} \circ \hat{\varphi}_{com_1} \circ \hat{\sigma}_1 \circ \varphi_{chl_1}$  is trivial with non-negligible probability. This deduces that  $\Pr[[\sigma_1]_*(\phi_2 \circ \hat{\phi}_1) = \varphi_2 \circ \hat{\varphi}_1]$  is also non-negligible.

Take an honest prover as input in Algorithm 2. From Lemmas 1 and 2, we know that  $\Pr[[\sigma_1]_*(\phi_2 \circ \hat{\phi}_1) = \varphi_2 \circ \hat{\varphi}_1] < (f' + 1)(\ell'^{-2f}) \approx 2^{-2\lambda}$ , where  $\phi_1, \phi_2, \varphi_1$  and  $\varphi_2$  are chosen uniformly at random. This is a contradiction, as  $\Pr[[\sigma_1]_*(\phi_2 \circ \hat{\phi}_1) = \varphi_2 \circ \hat{\varphi}_1]$  is non-negligible.

Take a malicious prover as input in Algorithm 2. Same as above, the malicious prover has to ensure  $[\sigma_1]_*(\phi_2 \circ \hat{\phi}_1) = \varphi_2 \circ \hat{\varphi}_1$ . Note that  $[\sigma_1]_*(\hat{\phi}_1) = \hat{\varphi}_1$  is a necessary condition for  $[\sigma_1]_*(\phi_2 \circ \hat{\phi}_1) = \varphi_2 \circ \hat{\varphi}_1$ . Adapting Lemma 3, ensuring  $[\sigma_1]_*(\hat{\phi}_1) = \hat{\varphi}_1$  is easy for the prover. However, since  $\varphi_2$  and  $\phi_2$  are chosen uniformly at random after  $\sigma_1$  is generated,

$$\Pr \left[ [\sigma_1]_*(\phi_2 \circ \hat{\phi}_1) = \varphi_2 \circ \hat{\varphi}_1 \mid [\sigma_1]_*(\hat{\phi}_1) = \hat{\varphi}_1 \right] = (\ell' + 1)^{-1} (\ell')^{-f'+1} \approx 2^{-\lambda},$$

which is still negligible. This contradicts with the assumption.  $\square$

## 4.2 Zero Knowledge

Same as the security proof for the SQIsignHD identification protocol, we use the following oracle to prove the PIsignHD identification protocol is special honest verifier zero-knowledge.

**Definition 2 ([11, Definition 20]).** *A random uniform good degree isogeny oracle (RUGDIO) is an oracle taking as input a supersingular curve  $E/\mathbb{F}_{p^2}$  and returning an efficient representation  $(\sigma(P_1), \sigma(Q_1), q)$  of a random isogeny  $\sigma : E \rightarrow E'$ , where  $\{P_1, Q_1\}$  is a canonical basis of  $E[\ell^f]$  and  $q$  is the degree of  $\sigma$  which is  $\ell^g$ -good and coprime to  $\ell'$ . Besides, the RUGDIO model satisfies that*

- *The distribution of  $E'$  is uniform in the supersingular isogeny graph.*
- *The conditional distribution of  $\sigma$  given  $E$  is uniform among isogenies from  $E$  to  $E'$  of  $\ell^g$ -good degree coprime to  $\ell'$ .*

With the help of the RUGDIO model, one can generate an efficient representation of an isogeny starting from a given supersingular elliptic curve  $E_1$ , whose degree is  $\ell^g$ -good degree coprime to  $\ell'$ . It has been argued in [11, Section 5.3] that access to the oracle does not offer any advantage in reducing the hardness of *Supersingular Endomorphism Ring Problem* (Problem 2), which can be reduced to *Supersingular Endomorphism Problem* (Problem 1) [18,30]. Same as the SQIsignHD identification protocol, we also have a heuristic assumption on the distribution of the commitment  $E_1$ .

*Problem 2 (Supersingular Endomorphism Ring Problem).* Given a prime  $p$  and a supersingular elliptic curve  $E$  defined over  $\mathbb{F}_{p^2}$ , find four endomorphisms of  $E$  that can be efficiently evaluated, to form a basis of the endomorphism ring of  $E$ .

**Proposition 3.** *Assume that the commitment  $E_1$  is computationally indistinguishable from an elliptic curve chosen uniformly at random in the supersingular isogeny graph. Then the PIsignHD identification protocol is special honest verifier zero-knowledge in the RUGDIO model. In other words, there exists a simulator  $\mathcal{S}$  with access to RUGDIO, satisfying that the distribution of the accepting conversation generated by  $\mathcal{S}$  is computationally indistinguishable from the conversation of the PIsignHD identification protocol.*

*Proof.* We proceed similarly as the zero-knowledge proof of SQIsignHD [11, Theorem 21]. The simulator  $\mathcal{S}$  is constructed as follows: Firstly, the simulator  $\mathcal{S}$  selects an  $\ell'^{f'}$ -isogeny  $\varphi'_{chl} : E_A \rightarrow E'_3$  uniformly at random. After that, the simulator adapts the RUGDIO model to generate an efficient representation  $R'$  of  $\hat{\sigma}'$  from  $E'_3$  to  $E'_2$ , which is also an efficient representation of  $\sigma$  from  $E'_2$  to  $E'_3$ . Finally, the simulator  $\mathcal{S}$  generates an  $\ell'^{f'}$ -isogeny  $\hat{\varphi}'_{com} : E'_2 \rightarrow E'_1$  uniformly at random. The conversation of  $\mathcal{S}$  is of form  $(E'_1, \varphi'_{com}, R', \varphi'_{chl})$ .

Assuming that the conversation of the PIsignHD identification protocol is of form  $(E_1, \varphi_{com}, R, \varphi_{chl})$ , we aim to prove that the distribution of  $(E'_1, \varphi'_{com}, R', \varphi'_{chl})$  is computationally indistinguishable from that of  $(E_1, \varphi_{com}, R, \varphi_{chl})$ . Applying the RUGDIO model, the curve  $E'_2$  is chosen uniformly at random in the supersingular isogeny graph. Since the isogeny  $\hat{\varphi}'_{com} : E'_2 \rightarrow E'_1$  is also chosen

uniformly at random, it follows that  $E'_1$  is computationally indistinguishable from an elliptic curve chosen uniformly at random in the supersingular isogeny graph. Therefore,  $E_1$  and  $E'_1$  have the same distribution. Furthermore, the isogenies  $\varphi_{com}$  and  $\varphi'_{com}$  start from elliptic curves chosen uniformly at random in the supersingular isogeny graph and they are chosen uniformly at random, thus  $\varphi_{com}$  and  $\varphi'_{com}$  have the same distribution. Since  $\varphi_{chl} : E_A \rightarrow E_3$  and  $\varphi'_{chl} : E_A \rightarrow E'_3$  are chosen uniformly at random, thus they are indistinguishable.

Now we prove the efficient representations of  $\sigma$  and  $\sigma'$  are indistinguishable. From the second property of the RUGDIO model, the conditional distribution of  $\hat{\sigma}'$  given  $E'_3$  is uniform among isogeny from  $E'_3$  to  $E'_2$ , i.e., the conditional distribution of  $\sigma'$  given  $E'_2$  is uniform among isogeny from  $E'_2$  to  $E'_3$ . From [11, Section 4.2],  $\sigma$  has the same distribution conditionally to  $E_2$  and  $E_3$ . Notably,  $E_2$ ,  $E'_2$ ,  $E_3$ ,  $E'_3$  are computationally indistinguishable from an elliptic curve chosen uniformly at random in the supersingular isogeny graph. This ends the proof.  $\square$

## 5 Implementation and Comparison

In this section, we show how to further reduce the signature size of PSignHD thanks to the public storage (or the storage of the verifier's size). Besides, we explore how to implement PSignHD with fast online signing via offline computations, and report the online/offline signature performance results of PSignHD. Comparisons between SQSignHD and PSignHD are also discussed in detail.

### 5.1 Signature compactness

Recall that the signature of PSignHD is of the form  $(E_1, R)$  where  $R = (\sigma(P_2), \sigma(Q_2), q)$ : the domain  $E_1$  of the isogeny  $\varphi_{com}$ , the evaluation on the canonical basis  $\{P_2, Q_2\}$  of  $E_2[\ell^f]$  through  $\sigma$  and the degree of  $\sigma$ . The size is the same as that of SQSignHD.

Indeed, the signer can also transmit  $(E_2, R, \ker(\hat{\varphi}_{com}))$  as the signature: the codomain  $E_2$  of the isogeny  $\varphi_{com}$ , the evaluation on the canonical basis  $\{P_2, Q_2\}$  of  $E_2[\ell^f]$  through  $\sigma$ , the degree of  $\sigma$  and the kernel of  $\hat{\varphi}_{com}$ . In this scenario, the signature involves the additional information  $\ker(\hat{\varphi}_{com})$ . Therefore, the signature size is larger than that of SQSignHD. In the following, we show how to compress  $(E_2, R, \ker(\hat{\varphi}_{com}))$ , making it more compact than the SQSignHD signature.

Similar to SQSignHD, one can also compress the torsion basis information utilizing the technique in Appendix C. Let  $\{P_0, Q_0\}$ ,  $\{P_1, Q_1\}$ ,  $\{P_2, Q_2\}$  and  $\{P_A, Q_A\}$  be the canonical bases of  $E_0[\ell^f]$ ,  $E_1[\ell^f]$ ,  $E_2[\ell^f]$  and  $E_A[\ell^f]$ , respectively. Assume that

$$\begin{aligned} \begin{pmatrix} P_A \\ Q_A \end{pmatrix} &= M_\tau \begin{pmatrix} \tau(P_0) \\ \tau(Q_0) \end{pmatrix}, & \hat{\gamma} \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix} &= M_{\hat{\gamma}} \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix}, \\ \psi \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix} &= M_\psi \begin{pmatrix} P_1 \\ Q_1 \end{pmatrix}, & \varphi_{com} \begin{pmatrix} P_1 \\ Q_1 \end{pmatrix} &= M_{\varphi_{com}} \begin{pmatrix} P_2 \\ Q_2 \end{pmatrix}. \end{aligned} \tag{2}$$

where  $M_\tau, M_{\hat{\gamma}}, M_\psi, M_{\varphi_{com}} \in \mathbb{M}_2(\mathbb{Z}/\ell^f \mathbb{Z})$ . Note that

$$\sigma = \frac{\varphi_{chl} \circ \tau \circ \gamma \circ \hat{\psi} \circ \hat{\varphi}_{com}}{[\deg(\varphi_{chl}) \deg(\tau) \deg(\psi) \deg(\varphi_{com})]}.$$

Then

$$\begin{aligned} \hat{\sigma} \circ \varphi_{chl} \circ \tau \left( \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix} \right) &= \frac{\varphi_{com} \circ \psi \circ \hat{\gamma} \circ \hat{\tau} \circ \hat{\varphi}_{chl} \circ \varphi_{chl} \circ \tau}{[\deg(\varphi_{chl}) \deg(\tau) \deg(\psi) \deg(\varphi_{com})]} \left( \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix} \right) \\ &= \frac{\varphi_{com} \circ \psi \circ \hat{\gamma}}{[\deg(\psi) \deg(\varphi_{com})]} \left( \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix} \right). \end{aligned}$$

It follows from  $\hat{\gamma} \left( \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix} \right) = M_{\hat{\gamma}} \left( \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix} \right)$  that

$$\hat{\sigma} \circ \varphi_{chl} \circ \tau \left( \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix} \right) = \frac{M_{\hat{\gamma}}}{[\deg(\psi) \deg(\varphi_{com})]} \cdot \varphi_{com} \circ \psi \left( \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix} \right).$$

Further, from  $\psi \left( \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix} \right) = M_\psi \left( \begin{pmatrix} P_1 \\ Q_1 \end{pmatrix} \right)$  and  $\varphi_{com} \left( \begin{pmatrix} P_1 \\ Q_1 \end{pmatrix} \right) = M_{\varphi_{com}} \left( \begin{pmatrix} P_2 \\ Q_2 \end{pmatrix} \right)$ ,

$$\hat{\sigma} \circ \varphi_{chl} \circ \tau \left( \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix} \right) = \frac{M_{\hat{\gamma}} \cdot M_\psi}{[\deg(\psi) \deg(\varphi_{com})]} \cdot \varphi_{com} \left( \begin{pmatrix} P_1 \\ Q_1 \end{pmatrix} \right) = \frac{M_{\hat{\gamma}} \cdot M_\psi \cdot M_{\varphi_{com}}}{[\deg(\psi) \deg(\varphi_{com})]} \left( \begin{pmatrix} P_2 \\ Q_2 \end{pmatrix} \right).$$

Same as the deduction in Equation (4):

$$M_\tau \cdot \begin{pmatrix} \hat{\sigma} \circ \varphi_{chl}(\tau(P_0)) \\ \hat{\sigma} \circ \varphi_{chl}(\tau(Q_0)) \end{pmatrix} = \hat{\sigma} \circ \varphi_{chl} \left( M_\tau \cdot \begin{pmatrix} \tau(P_0) \\ \tau(Q_0) \end{pmatrix} \right) = \hat{\sigma} \circ \varphi_{chl} \left( \begin{pmatrix} P_A \\ Q_A \end{pmatrix} \right).$$

As a consequence,

$$\hat{\sigma} \circ \varphi_{chl} \left( \begin{pmatrix} P_A \\ Q_A \end{pmatrix} \right) = \frac{M_\tau \cdot M_{\hat{\gamma}} \cdot M_\psi \cdot M_{\varphi_{com}}}{[\deg(\psi) \deg(\varphi_{com})]} \left( \begin{pmatrix} P_2 \\ Q_2 \end{pmatrix} \right).$$

Therefore, the signature can be compressed into  $(E_2, M, q, \ker(\hat{\varphi}_{com}))$ , where

$$M = \frac{M_\tau \cdot M_{\hat{\gamma}} \cdot M_\psi \cdot M_{\varphi_{com}}}{[\deg(\psi) \deg(\varphi_{com})]} = \frac{M_\tau \cdot M_{\hat{\gamma}} \cdot M_\psi \cdot M_{\varphi_{com}}}{[\deg(\psi) \ell'^f]}.$$

To store  $\ker(\hat{\varphi}_{com})$ , we compress it by finding  $k_{\hat{\varphi}_{com}} \in \mathbb{Z}/\ell'^f \mathbb{Z}$  such that  $\ker(\hat{\varphi}_{com})$  can be represented by  $\langle P'_2 + [k_{\hat{\varphi}_{com}}]Q'_2 \rangle$  or  $\langle Q'_2 + [k_{\hat{\varphi}_{com}}]P'_2 \rangle$  where  $\{P'_2, Q'_2\}$  is the canonical basis of  $E_2[\ell'^f]$ . Note that  $\{P'_2, Q'_2\}$  can be recovered as  $E_2$  is given. Therefore, one can transfer  $(k_{\hat{\varphi}_{com}}, \text{label}_{\hat{\varphi}_{com}})$  instead of a generator of  $\ker(\hat{\varphi}_{com})$ , where  $\text{label}_{\hat{\varphi}_{com}}$  is a bit used to distinguish the two cases mentioned above. This reduces the storage cost of  $\ker(\hat{\varphi}_{com})$  to approximately  $\lambda$  bits.

As a  $\Gamma$ -signature, PIsignHD allows the signer to precompute all the intermediate values which are irrelevant to the message. In particular, the signer can precompute plenty of commitments, and store a list of codomains of the hash



isogenies  $D = \{E_2^{(1)}, E_2^{(2)}, \dots, E_2^{(n)}\}$  in public, or on the verifier's side. Hence, the signer can transfer the index  $ind_E$  in  $D$  instead of the codomain of the hash isogeny. Generally, setting  $n = 2^{32}$  is enough for practice.

With the help of the list  $D$ , the signature of PIsignHD can be compressed into  $(ind_E, M, q, (k_{\hat{\varphi}_{com}}, label_{\hat{\varphi}_{com}}))$ . From Remark 4, the entire action matrix  $M$  can be recovered once three entries of it are known, and its size can be further halved by revealing the actions of  $\sigma$  on a  $2^{\lceil g/2 \rceil}$ -torsion basis instead of  $\{P_2, Q_2\}$ . Therefore, the total storage cost is approximately  $32 + (3 \cdot 0.5\lambda + 1) + \lambda + (\lambda + 1) = (3.5\lambda + 34)$  bits. For comparison, the signature size of SQIsignHD is about  $6.5\lambda$  bits. For NIST-I security level ( $\lambda = 128$ ), the signature size of PIsignHD is 519 bits, while the storage cost of SQIsignHD is 870 bits.

## 5.2 Offline/online signatures

As mentioned in Section 2.2, the isogeny  $\varphi_{chl}$  can be recovered by the verifier, and the signer can avoid isogeny computations relevant to  $\varphi_{chl}$ . Besides, the isogenies  $\tau$  and  $\tau'$  have been constructed in key generation. Therefore, the bottleneck of the response in SQIsignHD is the isogeny computations of the commitment.

In PIsignHD, the signer not only computes the codomain of  $\psi$  but an equivalent isogeny  $\psi'$  of coprime degree, due to the translation from the isogeny  $\varphi_{com}$  to the associated ideal  $I_{\varphi_{com}}$ . In addition, the signer has to construct and evaluate the isogeny  $\varphi_{com}$  to obtain the codomain  $E_2$  and the action matrix  $M_\psi$  associated to  $\psi$ . Fortunately, by implementing online/offline computations, all the above parts can be computed offline and thus they do not affect the efficiency of the online response. Detailed descriptions of the offline/online signatures are presented in Algorithms 3 and 4.

*Remark 3.* The signer can compute  $\frac{M_{\varphi_{com} \circ \psi}}{\deg(\psi)^{\ell' f'}}$  offline and store it instead of  $M_{\varphi_{com} \circ \psi}$  to further improve the online signing (Step 6 of Algorithm 4).

The constructions of  $\psi$  and  $\psi'$  are the efficiency bottlenecks of the offline computations. There are mainly two methods to achieve this: One is to generate  $\psi$  uniformly at random, then compute the associated ideal  $I_\psi$  and apply the KLPT algorithm [24] to obtain an equivalent ideal, and finally translate it to the associated isogeny  $\psi'$  (note that in this case the degree of  $\psi'$  is approximately  $p^3$ ); the other is to generate both of them simultaneously by the elegant techniques utilized in the key generation phase of SQIsignHD [11, Section 3.3]. Our implementation applies the latter one for efficiency. To reduce the storage cost for the ideal  $I_{\varphi_{com}} \cdot I_\psi$ , the signer can execute the algorithm **RandomEquivalentIdeal** <sub>$\ell_g$</sub>  to generate an ideal  $I \sim I_{\varphi_{com}} \cdot I_\psi$  with norm  $\text{Nrd}(I) \approx \sqrt{p}$ . Note that the codomain  $E_2$  can be public or stored on the verifier's side, while the tuple  $(M_{\varphi_{com} \circ \psi}, I, k_{\hat{\varphi}_{com}}, label_{\hat{\varphi}_{com}})$  should be secret.

The online signature avoids the isogeny computations, thus all the operations are over the quaternions and linear algebra. Particularly, the efficiency bottleneck is the generation of the ideal associated to  $\sigma$ . Currently, the approach to obtain the target ideal is somewhat primitive. Finding a more efficient method for the

---

**Algorithm 3** Offlinesignature

---

**Require:** The initial curve  $E_0$  with known Endomorphism ring.

**Ensure:** The curve  $E_2$ , the action matrix  $M_{\varphi_{com} \circ \psi} = M_{\varphi_{com}} \cdot M_{\psi}$  with  $M_{\varphi_{com}}$  and  $M_{\psi}$  defined in Equation (2), the ideal  $I$  associated to the isogeny  $\varphi_{com} \circ \psi : E_0 \rightarrow E_2$ , the integer  $k_{\hat{\varphi}_{com}}$  and a bit  $label_{\hat{\varphi}_{com}}$  used to determine  $\ker(\hat{\varphi}_{com})$ .

- 1: Generate a random isogeny walk  $\psi : E_0 \rightarrow E_1$  of degree  $\ell^{\bullet} \approx p$  and an equivalent isogeny  $\psi' : E_0 \rightarrow E_1$  of degree  $\ell^{\bullet} \approx p$ ;
  - 2: Compute the ideals  $I_{\psi}$  and  $I_{\psi'}$  associated to  $\psi$  and  $\psi'$ , respectively;
  - 3: Compute the canonical bases  $\{P_0, Q_0\}$  and  $\{P_1, Q_1\}$  of  $E_0[\ell^f]$  and  $E_1[\ell^f]$ , respectively;
  - 4: Compute the action matrix  $M_{\psi}$  as defined in Equation (2);
  - 5:  $\varphi_{com} \leftarrow \mathcal{H}'(E_1, \mathcal{H}(j(E_1)))$  and compute  $\varphi_{com}(P_1)$  and  $\varphi_{com}(Q_1)$ ;
  - 6: Compute  $\ker(\hat{\varphi}_{com})$ , the kernel of  $\hat{\varphi}_{com}$ ;
  - 7:  $I_{\varphi_{com}} \leftarrow \mathbf{IsogenyToIdeal}(\varphi_{com}, \psi', I_{\psi'})$ ;
  - 8: Compute the canonical basis  $\{P'_2, Q'_2\}$  of  $E_2[\ell^{f'}]$ ;
  - 9: Compute the action matrix  $M_{\varphi_{com}}$  as defined in Equation (2);
  - 10:  $M_{\varphi_{com} \circ \psi} \leftarrow M_{\varphi_{com}} \cdot M_{\psi}$ ,  $I \leftarrow I_{\varphi_{com}} \cdot I_{\psi}$ ;
  - 11: Find  $k_{\hat{\varphi}_{com}} \in \mathbb{Z}/\ell^{f'}\mathbb{Z}$  such that  $\ker(\hat{\varphi}_{com}) = \langle P'_2 + [k_{\hat{\varphi}_{com}}]Q'_2 \rangle$  or  $\ker(\hat{\varphi}_{com}) = \langle [k_{\hat{\varphi}_{com}}]P'_2 + Q'_2 \rangle$ ;
  - 12:  $label_{\hat{\varphi}_{com}} \leftarrow 1$  if  $\ker(\hat{\varphi}_{com}) = \langle P'_2 + [k_{\hat{\varphi}_{com}}]Q'_2 \rangle$ , or  $label_{\hat{\varphi}_{com}} \leftarrow 0$  otherwise;
  - 13:  $I \leftarrow \mathbf{RandomEquivalentIdeal}_{\ell^g}(I)$ ;
  - 14: **return**  $E_2$  and  $(M_{\varphi_{com} \circ \psi}, I, k_{\hat{\varphi}_{com}}, label_{\hat{\varphi}_{com}})$ .
- 

$\ell^g$ -good equivalent ideal generation is essential to improve the performance of the online signature. We leave it as future work.

We note that the online signing phase of SQIsignHD can also be accelerated via precomputation. Precisely, the signer can precompute the isogeny  $\psi$ , the codomain  $E_1$  and the action matrices such as  $M_{\psi}$  in Equation (3). This also avoids the isogeny computations in the online signing phase. However, SQIsignHD has several disadvantages in applications compared with PIsignHD when applying the offline/online computations:

- SQIsignHD requires larger storage requirements. To generate a signature with respect to the commitment  $E_1$ , the signer has to store  $I_{\psi}$ ,  $M_{\psi}$  and  $E_1$ . Especially, since the commitment  $E_1$  cannot be public, the signer has to store it before signing. For comparison, PIsignHD allows the list  $D = \{E_2^{(1)}, E_2^{(2)}, \dots, E_2^{(s)}\}$  to be public or be stored on the verifier's side. As a consequence, the signer only stores the information  $I \sim I_{\varphi_{com}} \cdot I_{\psi}$ ,  $M_{\varphi_{com} \circ \psi}$ ,  $(k_{\hat{\varphi}_{com}}, label_{\hat{\varphi}_{com}})$  and a label  $ind_E$  instead when implementing PIsignHD, reducing the storage cost by approximately  $3\lambda$  bits for each commitment.
- SQIsignHD has larger signature size. As discussed previously, PIsignHD allows the signer to further compress the signature thanks to the public list  $D$ . On the other hand, in SQIsignHD the knowledge of  $E_1$  should be entirely transferred as it is not allowed to be public in advance. Although the signature of PIsignHD also involves the knowledge of the hash isogeny  $\varphi_{com}$ ,

---

**Algorithm 4** Onlinesignature

---

**Require:** The isogeny  $\tau' : E_0 \rightarrow E_A$  of degree  $\ell'^{\bullet} \approx p$ , the ideals  $I_\tau$  and  $I_{\tau'}$  associated to  $\tau$  and  $\tau'$  respectively, the ideal  $I_{\varphi_{com} \circ \psi}$  equivalent to  $I_{\varphi_{com}} \cdot I_\psi$ , the isogeny  $\varphi_{chl} : E_A \rightarrow E_2$  of degree  $\ell'^{f'}$ , and the action matrices  $M_\tau$  and  $M_{\varphi_{com} \circ \psi}$  defined in Equation (2).

**Ensure:** The matrix  $M$  such that  $(\hat{\sigma} \circ \varphi_{chl}(P_A), \hat{\sigma} \circ \varphi_{chl}(Q_A))^T = M \cdot (P_1, Q_1)^T$  and the degree  $q$  of the isogeny  $\sigma : E_1 \rightarrow E_2$ .

- 1:  $I_{\varphi_{chl}} \leftarrow \mathbf{IsogenyToIdeal}(\varphi_{chl}, \tau', I_{\tau'}), J \leftarrow \overline{I_{\varphi_{com} \circ \psi}} \cdot I_\tau \cdot I_{\varphi_{chl}};$
  - 2:  $I \leftarrow \mathbf{RandomEquivalentIdeal}_{\ell_1^g}(J)$  and compute the reduced norm  $q$  of  $I$ ;
  - 3: If  $q$  is not  $\ell^g$ -good or  $\gcd(q, \ell') \neq 1$ , go back to Line 2;
  - 4: Compute  $\gamma \in \mathcal{O}$  such that  $\mathcal{O}\gamma = I_\psi \cdot I \cdot \overline{I_\tau} \cdot \overline{I_{\varphi_{chl}}};$
  - 5: Compute the action matrix  $M_\gamma$  as defined in Equation (2);
  - 6:  $M \leftarrow \frac{M_\tau \cdot M_\gamma \cdot M_{\varphi_{com} \circ \psi}}{\deg(\psi) \ell'^{f'}};$
  - 7: **return**  $(M, q).$
- 

it is still more compact than that of SQIsignHD due to the large storage requirement of the curve coefficient.

- The challenge isogeny in SQIsignHD is generated from the knowledge of both the commitment  $E_1$  and the message  $m$ . Therefore, the online phase in SQIsignHD has to compute the hash of  $E_1$  and  $m$ , i.e.,  $\mathcal{H}(E_1||m)$ , and then generates the challenge isogeny  $\varphi = \mathcal{H}'(E_A, \mathcal{H}(E_1||m))$ . Conversely, in PIsignHD the challenge isogeny  $\varphi_{chl} = \mathcal{H}'(E_A, \mathcal{H}(m))$ . This is preferred in some specific applications. For example, the hash of the message  $m$  can be hashed by a trusted party. In this case, the signer is able to use a low-power device to generate the signature with respect to  $\mathcal{H}(m)$ , without handling the entire message. When applying SQIsignHD, the signer has to compute  $\mathcal{H}(E_1||m)$  or transmit  $E_1$  to the trusted party. The former enlarges the communication cost and the computational cost of online signing, while the latter requires an additional round of interaction.
- In PIsignHD, the verifier can precompute some intermediate values to fasten the verification. More details are left in the next subsection.

### 5.3 Experimental Results

Based on the SQIsignHD code <sup>3</sup>, we implement the online/offline signatures of PIsignHD. We benchmark our code on Intel(R) Core(TM) i9-12900K 3.20 GHz with TurboBoost and hyperthreading features disabled. The code is available at

<https://github.com/Kaizhan-Lin/PIsignHD>.

As mentioned in this last subsection, SQIsignHD also benefits from the online/offline computations. In Table 1 we give an efficiency comparison between the offline/online responses of SQIsignHD and PIsignHD.

---

<sup>3</sup> <https://github.com/Pierrick-Dartois/SQIsignHD-lib>

Table 1: Comparison of the SQIsignHD and PIsignHD signing implementations targeting the NIST-I security level. For the performance results (expressed in millions of clock cycles), we execute 1000 times for a 256-bit message and record the average time.

| Implementation                    |                        | SQIsignHD | PIsignHD |
|-----------------------------------|------------------------|-----------|----------|
| Signature size (bits)             | Original               | 870       | 870      |
|                                   | Compressed             | -         | 519      |
| Clock cycles ( $cc \times 10^6$ ) | Original               | 70.1      | 89.8     |
|                                   | Offline (Uncompressed) | 57.9      | 77.8     |
|                                   | Online (Uncompressed)  | 12.0      | 11.8     |
|                                   | Offline (Compressed)   | -         | 89.6     |
|                                   | Online (Compressed)    | -         | 11.8     |

As expected, the online response of SQIsignHD and PIsignHD is very fast and close. According to our experimental results, the online response takes only 4 ms. For comparison, the signature of SQIsignHD takes 22.4 ms on average. Therefore, the online response of SQIsignHD/PIsignHD is over 5 times faster than the entire signing procedure of SQIsignHD without offline precomputations.

While the implementation efficiency of online responses of PIsignHD remains unchanged regardless of whether the signature compression is employed, the offline computation is less efficient in the case when using the compression technique. The main reason is that the signer needs to compute the kernel of  $\hat{\varphi}_{com}$  and compress it to  $(k_{\hat{\varphi}_{com}}, label_{\hat{\varphi}_{com}})$  by computing discrete logarithms during the offline phase of compressed PIsignHD.

In our implementation, we improve the performance of discrete logarithms in the signing phase by utilizing reduced Tate pairings [25]. Indeed, there are some other techniques in the literature which can be utilized to improve the implementation of the offline computations. For instance, one can employ interleaved modular multiplication algorithms [26] to reduce considerable memory loads and stores for multiplications in  $\mathbb{F}_{p^2}$ . Very recently, faster approaches for pairing computations in isogeny-based protocols are explored by [34,5], which are particularly beneficial for the acceleration of the action matrix computations. We note that in the real-world applications, the offline computations can be connected to the power. Hence, it is acceptable that the offline computations of PIsignHD are not as efficient as that of SQIsignHD.

In summary, the online response performance of both signatures is very close, while SQIsignHD has a faster implementation of the offline computations compared to PIsignHD. However, regarding various advantages as discussed in Section 5.2, PIsignHD appears more promising in practical applications.

Now we analyze the performance of other parts in PIsignHD.

The key generation phase of PIsignHD is identical to that of SQIsignHD, and thus the performance is the same.

When we do not apply the online/offline technique, the verification in PIsignHD needs to construct  $\varphi_{com} : E_1 \rightarrow E_2$ , which is the hash of  $E_1$ . Since  $\varphi_{com}$  is a

power-smooth isogeny that can be efficiently constructed and evaluated, the overhead is negligible as the isogeny computations in high dimension dominate the computational cost. Therefore, the verification performance of PIsignHD is very close to that of SQIsignHD.

When adapting the online/offline technique, PIsignHD has the potential to achieve a better verification performance compared to SQIsignHD. In addition, some intermediate values can also be precomputed to fasten the verification. For example, the verifier can precompute the canonical basis of any supersingular curve in the list  $D$ . Besides, as the challenge isogeny can be generated without any interaction with the signer, the verifier can also compute the canonical basis of the codomain  $E_3$  in advance. We are confident that the isogeny computation in high dimension can be accelerated via precomputation with further research.

## 6 Conclusion

In this paper we introduced a new structure for the SQIsign family, and proposed PIsignHD based on SQIsignHD. The flexible challenge generation benefits the implementation of PIsignHD in the real-world applications. Furthermore, PIsignHD has a shorter signature size compared with SQIsignHD. In addition, PIsignHD achieves a fast online response via offline computations with cheaper storage requirements.

In our future work, we aim to further enhance the performance of PIsignHD, including reducing the offline storage complexity for the prover, improving the efficiency of offline/online signing and verification, etc. We will also adapt the new structure to other efficient variants of SQIsign [10,1,28,17] to make them more competitive in applications. Additionally, it is interesting to develop practical  $\Gamma$ -signatures based on other isogeny-based protocols, such as CSIDH [7] and SIDH-like schemes [22,2].

## Acknowledgment

We thank Yi-Fu Lai and all the anonymous reviewers for their constructive suggestions. Kaizhan Lin, Weize Wang and Yunlei Zhao are supported by The National Key Research and Development Program of China (No.2022YFB2701601), Shanghai Collaborative Innovation Fund (No.XTCX-KJ-2023-54) and Special Fund for Key Technologies in Blockchain of Shanghai Scientific and Technological Committee (No.23511100300). Chang-An Zhao is supported by the National Natural Science Foundation of China (No. 12441107), Guangdong Basic and Applied Basic Research Foundation (No. 2025A1515011764), Guangdong Major Project of Basic and Applied Basic Research (No. 2019B030302008) and Guangdong Provincial Key Laboratory of Information Security Technology (No. 2023B1212060026). 2019B030302008.

## References

1. Basso, A., Dartois, P., Feo, L.D., Leroux, A., Maino, L., Pope, G., Robert, D., Wesolowski, B.: SQIsign2D–West. In: Chung, K.M., Sasaki, Y. (eds.) *Advances in Cryptology – ASIACRYPT 2024*. pp. 339–370. Springer Nature Singapore, Singapore (2025)
2. Basso, A., Fouotsa, T.B.: New SIDH Countermeasures for a More Efficient Key Exchange. In: Guo, J., Steinfeld, R. (eds.) *Advances in Cryptology – ASIACRYPT 2023*. pp. 208–233. Springer Nature Singapore, Singapore (2023)
3. Bernstein, D.J., de Feo, L., Leroux, A., Smith, B.: Faster computation of isogenies of large prime degree. In: Galbraith, S. (ed.) *ANTS-XIV - 14th Algorithmic Number Theory Symposium. Proceedings of the Fourteenth Algorithmic Number Theory Symposium (ANTS-XIV)*, vol. 4, pp. 39–55. Mathematical Sciences Publishers, Auckland, New Zealand (2020)
4. Beullens, W., Kleinjung, T., Vercauteren, F.: CSI-FiSh: Efficient Isogeny Based Signatures Through Class Group Computations. In: Galbraith, S.D., Moriai, S. (eds.) *Advances in Cryptology – ASIACRYPT 2019*. pp. 227–247. Springer International Publishing, Cham (2019)
5. Cai, S., Lin, K., Zhao, C.A.: Pairing Optimizations for Isogeny-Based Cryptosystems. *IET Information Security* **2024**(1), 9631360 (2024). <https://doi.org/https://doi.org/10.1049/2024/9631360>
6. Castryck, W., Decru, T.: An Efficient Key Recovery Attack on SIDH. In: *Advances in Cryptology–EUROCRYPT 2023: 42nd Annual International Conference on the Theory and Applications of Cryptographic Techniques*, Lyon, France, April 23–27, 2023, Proceedings, Part V. pp. 423–447. Springer (2023)
7. Castryck, W., Lange, T., Martindale, C., Panny, L., Renes, J.: CSIDH: an efficient post-quantum commutative group action. In: *Advances in Cryptology–ASIACRYPT 2018: 24th International Conference on the Theory and Application of Cryptology and Information Security*, Brisbane, QLD, Australia, December 2–6, 2018, Proceedings, Part III 24. pp. 395–427. Springer (2018)
8. Charles, D.X., Lauter, K.E., Goren, E.Z.: Cryptographic Hash Functions from Expander Graphs. *Journal of Cryptology* **22**(1), 93–113 (2009)
9. Chavez-Saab, J., Santos, M.C.R., Feo, L.D., Eriksen, J.K., Hess, B., Kohel, D., Leroux, A., Longa, P., Meyer, M., Panny, L., Patranabis, S., Petit, C., Henriques, F.R., Schaeffler, S., Wesolowski, B.: SQIsign (2023), manuscript available at <http://sqisign.org>
10. Corte-Real Santos, M., Eriksen, J.K., Meyer, M., Reijnders, K.: AprèsSQI: Extra Fast Verification for SQIsign Using Extension-Field Signing. In: Joye, M., Leander, G. (eds.) *Advances in Cryptology – EUROCRYPT 2024*. pp. 63–93. Springer Nature Switzerland, Cham (2024)
11. Dartois, P., Leroux, A., Robert, D., Wesolowski, B.: SQISignHD: New Dimensions in Cryptography. *Cryptology ePrint Archive*, Paper 2023/436 (2023), <https://eprint.iacr.org/2023/436>
12. Dartois, P., Leroux, A., Robert, D., Wesolowski, B.: SQISignHD: New Dimensions in Cryptography. In: Joye, M., Leander, G. (eds.) *Advances in Cryptology – EUROCRYPT 2024*. pp. 3–32. Springer Nature Switzerland, Cham (2024)
13. De Feo, L., Galbraith, S.D.: SeaSign: Compact Isogeny Signatures from Class Group Actions. In: Ishai, Y., Rijmen, V. (eds.) *Advances in Cryptology – EUROCRYPT 2019*. pp. 759–789. Springer International Publishing, Cham (2019)

14. De Feo, L., Kohel, D., Leroux, A., Petit, C., Wesolowski, B.: SQISign: Compact Post-quantum Signatures from Quaternions and Isogenies. In: Moriai, S., Wang, H. (eds.) *Advances in Cryptology – ASIACRYPT 2020*. pp. 64–93. Springer International Publishing, Cham (2020)
15. De Feo, L., Leroux, A., Longa, P., Wesolowski, B.: New Algorithms for the Deuring Correspondence: Towards Practical and Secure SQISign Signatures. In: Hazay, C., Stam, M. (eds.) *Advances in Cryptology – EUROCRYPT 2023*. pp. 659–690. Springer Nature Switzerland, Cham (2023)
16. De Feo, L., Delpech de Saint Guilhem, C., Fouotsa, T.B., Kutas, P., Leroux, A., Petit, C., Silva, J., Wesolowski, B.: Séta: Supersingular Encryption from Torsion Attacks. In: Tibouchi, M., Wang, H. (eds.) *Advances in Cryptology – ASIACRYPT 2021*. pp. 249–278. Springer International Publishing, Cham (2021)
17. Duparc, M., Fouotsa, T.B.: SQIPrime: A Dimension 2 Variant of SQISignHD with Non-smooth Challenge Isogenies. In: Chung, K.M., Sasaki, Y. (eds.) *Advances in Cryptology – ASIACRYPT 2024*. pp. 396–429. Springer Nature Singapore, Singapore (2025)
18. Eisenträger, K., Hallgren, S., Lauter, K., Morrison, T., Petit, C.: Supersingular Isogeny Graphs and Endomorphism Rings: Reductions and Solutions. In: Nielsen, J.B., Rijmen, V. (eds.) *Advances in Cryptology – EUROCRYPT 2018*. pp. 329–368. Springer International Publishing, Cham (2018)
19. El Kaafarani, A., Katsumata, S., Pintore, F.: Lossy CSI-FiSh: Efficient Signature Scheme with Tight Reduction to Decisional CSIDH-512. In: Kiayias, A., Kohlweiss, M., Wallden, P., Zikas, V. (eds.) *Public-Key Cryptography – PKC 2020*. pp. 157–186. Springer International Publishing, Cham (2020)
20. Even, S., Goldreich, O., Micali, S.: On-Line/Off-Line Digital Signatures. In: Brassard, G. (ed.) *Advances in Cryptology — CRYPTO’ 89 Proceedings*. pp. 263–275. Springer New York, New York, NY (1990)
21. Fiat, A., Shamir, A.: How To Prove Yourself: Practical Solutions to Identification and Signature Problems. In: Odlyzko, A.M. (ed.) *Advances in Cryptology — CRYPTO’ 86*. pp. 186–194. Springer Berlin Heidelberg, Berlin, Heidelberg (1987)
22. Fouotsa, T.B., Moriya, T., Petit, C.: M-SIDH and MD-SIDH: Countering SIDH Attacks by Masking Information. In: Hazay, C., Stam, M. (eds.) *Advances in Cryptology – EUROCRYPT 2023*. pp. 282–309. Springer Nature Switzerland, Cham (2023)
23. Galbraith, S.D., Petit, C., Silva, J.: Identification Protocols and Signature Schemes Based on Supersingular Isogeny Problems. *Journal of Cryptology* **33**(1), 130–175 (Jan 2020)
24. Kohel, D., Lauter, K., Petit, C., Tignol, J.P.: On the quaternion  $\ell$ -isogeny path problem. *LMS Journal of Computation and Mathematics* **17**(A), 418–432 (2014)
25. Lin, K., Wang, W., Xu, Z., Zhao, C.A.: A Faster Software Implementation of SQISign. *IEEE Transactions on Information Theory* **70**(9), 6679–6689 (2024)
26. Longa, P.: Efficient Algorithms for Large Prime Characteristic Fields and Their Application to Bilinear Pairings. *IACR Transactions on Cryptographic Hardware and Embedded Systems* **2023**(3), 445–472 (2023)
27. Maino, L., Martindale, C., Panny, L., Pope, G., Wesolowski, B.: A direct key recovery attack on SIDH. In: *Advances in Cryptology–EUROCRYPT 2023: 42nd Annual International Conference on the Theory and Applications of Cryptographic Techniques*, Lyon, France, April 23–27, 2023, Proceedings, Part V. pp. 448–471. Springer (2023)

28. Nakagawa, K., Onuki, H., Castryck, W., Chen, M., Invernizzi, R., Lorenzon, G., Vercauteren, F.: SQIsign2D-East: A New Signature Scheme Using 2-Dimensional Isogenies. In: Chung, K.M., Sasaki, Y. (eds.) *Advances in Cryptology – ASIACRYPT 2024*. pp. 272–303. Springer Nature Singapore, Singapore (2025)
29. Onuki, H., Nakagawa, K.: Ideal-to-Isogeny Algorithm Using 2-Dimensional Isogenies and Its Application to SQIsign. In: Chung, K.M., Sasaki, Y. (eds.) *Advances in Cryptology – ASIACRYPT 2024*. pp. 243–271. Springer Nature Singapore, Singapore (2025)
30. Page, A., Wesolowski, B.: The Supersingular Endomorphism Ring and One Endomorphism Problems are Equivalent. In: Joye, M., Leander, G. (eds.) *Advances in Cryptology – EUROCRYPT 2024*. pp. 388–417. Springer Nature Switzerland, Cham (2024)
31. Pizer, A.K.: Ramanujan graphs and Hecke operators. *Bulletin of the American Mathematical Society* **23**(1), 127–137 (1990)
32. Renan, F., Kutas, P.: SQIAsignHD: SQIsignHD Adaptor Signature. *Cryptology ePrint Archive*, Paper 2024/561 (2024), <https://eprint.iacr.org/2024/561>
33. Robert, D.: Breaking SIDH in polynomial time. In: *Advances in Cryptology – EUROCRYPT 2023: 42nd Annual International Conference on the Theory and Applications of Cryptographic Techniques*, Lyon, France, April 23–27, 2023, *Proceedings*, Part V. pp. 472–503. Springer (2023)
34. Robert, D.: Fast pairings via biextensions and cubical arithmetic. *Cryptology ePrint Archive*, Paper 2024/517 (2024), <https://eprint.iacr.org/2024/517>
35. Silverman, J.H.: *The Arithmetic of Elliptic Curves*, 2nd Edition. *Graduate Texts in Mathematics*. Springer (2009)
36. Vélú, J.: Isogénies entre courbes elliptiques. *C. R. Acad. Sci., Paris, Sér. A* **273**, 238–241 (1971)
37. Voight, J.: *Quaternion algebras*. Springer Graduate Texts in Mathematics series. (2021)
38. Yao, A.C.C., Zhao, Y.: Online/Offline Signatures for Low-Power Devices. *IEEE Transactions on Information Forensics and Security* **8**(2), 283–294 (2013)



## A Mathematical Background

This section provides the necessary mathematical preliminaries, including elliptic curves, isogenies, quaternion algebras, orders and ideals. We refer to [35,37] for more details.

**Elliptic Curves.** Elliptic curves are nonsingular projective curves with genus 1. For applications, elliptic curves defined in this paper are over a finite field  $\mathbb{F}_q$ , denoted by  $E/\mathbb{F}_q$ , where  $q = p^n$  with prime  $p > 3$  and  $n \in \mathbb{N}^*$ . An isomorphism class of elliptic curves can be entirely determined by its  $j$ -invariant. We use  $j(E)$  to denote the  $j$ -invariant of  $E$ . All the rational points on the elliptic curve  $E$  and the point at infinity  $\infty_E^4$  form an abelian group  $E(\mathbb{F}_q)$  under point addition. Let  $\ell > 0$ , the  $\ell$ -torsion of  $E$  is defined as  $E[\ell] = \{P \in E(\overline{\mathbb{F}_q}) \mid [\ell]P = \infty_E\}$ , where  $[\ell]$  is a multiplication-by- $\ell$  map. An elliptic curve  $E$  is supersingular if  $E[p] = \{\infty_E\}$ , otherwise  $E$  is said to be ordinary.

**Isogenies.** An isogeny  $\varphi : E_1 \rightarrow E_2$  is a non-constant surjective morphism that sends  $\infty_{E_1}$  to  $\infty_{E_2}$ . Denote  $\deg(\varphi)$  the degree of  $\varphi$  as a rational map. Two curves  $E_1$  and  $E_2$  are said to be isogenous over  $\mathbb{F}_q$  if there exists an isogeny connecting them over  $\mathbb{F}_q$ . An isogeny  $\varphi$  is called cyclic if its kernel can be generated by one single point  $P$ , and separable if the cardinality of the kernel  $\ker(\varphi) = \{P \in E_1(\overline{\mathbb{F}_q}) \mid \varphi(P) = \infty_{E_2}\}$  is equal to  $\deg(\varphi)$ . If  $\deg(\varphi)$  is coprime to the characteristic of the finite field, then  $\varphi$  must be separable. We abbreviate a separable isogeny of degree  $\ell$  as an  $\ell$ -isogeny. Furthermore, for any isogeny  $\varphi : E_1 \rightarrow E_2$ , there exists a unique isogeny  $\hat{\varphi} : E_2 \rightarrow E_1$  such that  $\hat{\varphi} \circ \varphi = [\deg(\varphi)]$ , i.e., the composition of the two isogenies is a multiplication-by- $\deg(\varphi)$  map. In this case, we call  $\hat{\varphi}$  the dual isogeny of  $\varphi$ .

Let  $\varphi_1 : E_0 \rightarrow E_1$  and  $\varphi_2 : E_0 \rightarrow E_2$  be two separable isogenies with  $\gcd(\deg(\varphi_1), \deg(\varphi_2)) = 1$ . Then there exist two isogenies  $\psi_1 : E_2 \rightarrow E_3$  and  $\psi_2 : E_1 \rightarrow E_3$  such that  $\ker(\psi_1) = \varphi_2(\ker(\varphi_1))$  and  $\ker(\psi_2) = \varphi_1(\ker(\varphi_2))$ , as illustrated in Figure 6. We denote  $\psi_1 = [\varphi_2]_*\varphi_1$  (resp.  $\psi_2 = [\varphi_1]_*\varphi_2$ ) as the *pushforward* isogeny of  $\varphi_1$  (resp.  $\varphi_2$ ) through  $\varphi_2$  (resp.  $\varphi_1$ ). Conversely, the isogeny  $\varphi_1$  (resp.  $\varphi_2$ ) is called the *pullback* isogeny of  $\psi_1$  (resp.  $\psi_2$ ) through  $\varphi_2$  (resp.  $\varphi_1$ ), denoted by  $\varphi_1 = [\varphi_2]^*\psi_1$  (resp.  $\varphi_2 = [\varphi_1]^*\psi_2$ ). Note that  $\psi_1$  and  $\psi_2$  are also separable. In addition, there exists an isogeny  $\Phi : E_0 \rightarrow E_3$  such that  $\Phi = \psi_2 \circ \varphi_1 = \psi_1 \circ \varphi_2$ .

The supersingular  $\ell$ -isogeny graph is a graph whose vertices represent the supersingular  $\overline{\mathbb{F}_p}$  classes and edges represent the equivalent classes of  $\ell$ -isogenies connecting them. The graph is connected, essentially undirected and Ramanujan [31]. Moreover, the graph is  $\ell + 1$ -regular, meaning that there are exactly  $\ell + 1$  equivalent classes of isogenies starting from a given supersingular  $\overline{\mathbb{F}_p}$  class.

**Endomorphism rings.** An endomorphism of  $E$  is either an isogeny from  $E$  to itself, or the constant morphism  $[0]$ . The set of all the endomorphisms forms a ring under addition and composition, denoted by  $\text{End}(E)$ . The endomorphism

<sup>4</sup> The point at infinity of an elliptic curve is not necessarily the identity, but for simplicity we suppose that it is the identity point.

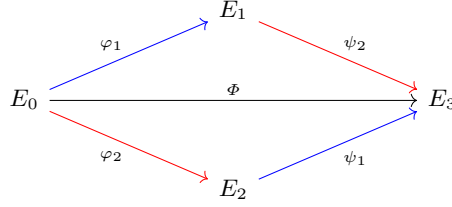


Fig. 6: A commutative isogeny diagram

ring  $\text{End}(E)$  is isomorphic to an order in a quaternion algebra if  $E$  is supersingular, or an order in a quadratic imaginary field if  $E$  is ordinary.

**Quaternion algebras, orders and ideals.** A quaternion algebra over  $\mathbb{Q}$  ramified at  $p$  and  $\infty$  has the form  $B_{p,\infty} = \mathbb{Q} + \mathbb{Q}i + \mathbb{Q}j + \mathbb{Q}k$ , where  $i^2 = -q$ ,  $j^2 = -p$  and  $k = ij = -ji$  with  $q \in \mathbb{Z}$ . The quaternion algebra has a canonical involution, mapping  $\alpha = \alpha_1 + \alpha_2 i + \alpha_3 j + \alpha_4 k$  to its conjugate  $\bar{\alpha} = \alpha_1 - \alpha_2 i - \alpha_3 j - \alpha_4 k$ . The reduced trace and the reduced norm of  $\alpha$  are defined as  $\text{Trd}(\alpha) = 2\alpha_1$  and  $\text{Nrd}(\alpha) = \alpha\bar{\alpha}$ , respectively.

An order in  $B_{p,\infty}$  is a full-rank lattice and also a subring. An order is called maximal if it is not contained in another order. A fractional ideal is a  $\mathbb{Z}$ -lattice of rank 4. Given an ideal  $I$ , its left order and right order are defined as

$$\mathcal{O}_L(I) = \{\alpha \in B_{p,\infty} | \alpha I \subset I\}, \mathcal{O}_R(I) = \{\alpha \in B_{p,\infty} | I\alpha \subset I\}.$$

A left (resp. right)  $\mathcal{O}$ -ideal  $I$  is a  $\mathbb{Z}$ -lattice of rank 4 satisfying that  $\mathcal{O} \subset \mathcal{O}_L(I)$  (resp.  $\mathcal{O} \subset \mathcal{O}_R(I)$ ) and  $\mathcal{O}_L(I)$  and  $\mathcal{O}_R(I)$  are maximal. An fractional ideal  $I$  is integral if  $I \subset \mathcal{O}_L(I)$ , which implies that  $I \subset \mathcal{O}_R(I)$ . Henceforth, we only focus on integral ideals and refer to them as ideals.

An ideal  $I$  is said to be invertible, if there exists an ideal  $I^{-1}$  such that  $II^{-1} = \mathcal{O}_L(I)$  or  $I^{-1}I = \mathcal{O}_R(I)$ . Denote  $\text{Nrd}(I) = \gcd\{\text{Nrd}(\alpha) | \alpha \in I\}$  the reduced norm of  $I$ , and  $\bar{I} = \{\bar{\alpha} | \alpha \in I\}$  the conjugate of  $I$ . If  $I$  is invertible, then  $I\bar{I} = \text{Nrd}(I)\mathcal{O}_L(I)$  and  $\bar{I}I = \text{Nrd}(I)\mathcal{O}_R(I)$ . An ideal  $I$  of integer reduced norm can be represented by  $I = \mathcal{O}_L(I)\alpha + \mathcal{O}_L(I)\text{Nrd}(I)$ , where  $\alpha \in \mathcal{O}_L(I)$ . Two left  $\mathcal{O}$ -ideals  $I$  and  $J$  are equivalent if there exists  $\beta \in B_{p,\infty}$  such that  $I = J\beta$ , denoted by  $I \sim J$ .

**Deuring correspondence.** The Deuring correspondence provides a link between the world of supersingular elliptic curves and the world of quaternion algebras.

Let  $E$  be a supersingular curve, and suppose that the endomorphism ring  $\text{End}(E)$  is isomorphic to a maximal order  $\mathcal{O}$  of  $B_{p,\infty}$ . Then an isogeny  $\varphi_I : E \rightarrow E'$  corresponds to a kernel ideal  $I = \{\alpha \in \mathcal{O} | \alpha(P) = \infty_E \text{ for all } P \in \ker(\varphi_I)\}$ , and  $\deg(\varphi_I) = \text{Nrd}(I)$ . Besides, the left order is isomorphic to  $\mathcal{O}$ , while the right order is isomorphic to  $\text{End}(E')$ . In particular, an endomorphism of  $E$  corresponds to a principal ideal. Conversely, given a left  $\mathcal{O}$ -ideal  $I$ , the kernel  $E[I] = \{P \in E(\overline{\mathbb{F}}_p) | \alpha(P) = \infty_E \text{ for all } \alpha \in I\}$  determines an isogeny  $\varphi_I$  with  $\ker(\varphi_I) = E[I]$  and  $\deg(\varphi_I) = \text{Nrd}(I)$ . The conjugation  $\bar{I}$  associates to

the dual isogeny  $\hat{\varphi}_I$ . The multiplication of ideals  $I \cdot J$  defines the composition  $\varphi_J \circ \varphi_I$ , where  $\varphi_I$  and  $\varphi_J$  are two isogenies associated to  $I$  and  $J$ , respectively. Note that in this case  $\mathcal{O}_R(I) \cong \mathcal{O}_L(J)$ . In addition, two left  $\mathcal{O}$ -ideals  $I$  and  $J$  are equivalent if and only if the isogenies  $\varphi_I$  and  $\varphi_J$  have the same domain and codomain up to isomorphism.

## B $\Sigma$ -Protocol and Fiat–Shamir Paradigm

Assume that  $\mathcal{P}$  and  $\mathcal{V}$  are probabilistic polynomial time machines, and the advantage of  $\mathcal{P}$  over  $\mathcal{V}$  is that  $\mathcal{P}$  knows  $w$  with  $(x, w) \in \mathcal{R}$ , where  $\mathcal{R}$  is an  $\mathcal{NP}$ -relation. Now concern the protocol that proceeds as follows:

- $\mathcal{P}$  sends a commitment  $a$  to  $\mathcal{V}$ ;
- $\mathcal{V}$  sends a random string  $e$  to  $\mathcal{P}$ ;
- $\mathcal{P}$  sends a reply  $z$  with respect to  $e$ , and  $\mathcal{V}$  accepts or rejects based on  $(x, a, e, z)$ .

**Definition 3.**  $\Sigma$ -protocol is a three-round public-coin protocol  $\langle \mathcal{P}, \mathcal{V} \rangle$  for an  $\mathcal{NP}$ -relation  $\mathcal{R}$  that proceeds as above. Besides,  $\Sigma$ -protocols should satisfy the following properties:

- **Completeness:**  $\mathcal{V}$  always accepts if  $\mathcal{P}$  and  $\mathcal{V}$  follow the protocol.
- **Special soundness:** Given two pairs of valid conversations  $(a, e, z)$  and  $(a, e', z')$  on any input  $x$  with  $e \neq e'$ , one can recover the witness  $w$  such that  $(x, w) \in \mathcal{R}$  in polynomial time with overwhelming probability.
- **Special honest verifier zero-knowledge (SHVZK):** There exists a probabilistic polynomial-time simulator  $\mathcal{S}$ , which takes as input  $x$ , and outputs an accepting conversation  $(a', e', z')$ , with the same (or computationally indistinguishable) probability distribution as the conversation  $(a, e, z)$  of the real protocol.

Given a  $\Sigma$ -protocol, Fiat–Shamir paradigm [21] can convert it to a signature scheme. The main idea is to set  $e = h(a||m)$ , where  $h$  is a hash function and  $m$  is the message. The modification allows the signer to sign the message without interacting with the verifier. The verifier accepts if  $(a, z)$  is a valid signature for  $m$ <sup>5</sup>.

## C Current Response Implementation of SQIsignHD

Suppose that  $\{P_0, Q_0\}$ ,  $\{P_1, Q_1\}$  and  $\{P_A, Q_A\}$  are the canonical bases of  $E_0[\ell^f]$ ,  $E_1[\ell^f]$  and  $E_A[\ell^f]$ , respectively. Then assume

$$\begin{pmatrix} P_A \\ Q_A \end{pmatrix} = M_\tau \begin{pmatrix} \tau(P_0) \\ \tau(Q_0) \end{pmatrix}, \hat{\gamma} \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix} = M_{\hat{\gamma}} \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix}, \psi \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix} = M_\psi \begin{pmatrix} P_1 \\ Q_1 \end{pmatrix}, \quad (3)$$

<sup>5</sup> In some specific signature schemes, such as SQIsign [9], the signature can be of form  $(e, z)$  since  $a$  can be recovered from  $(e, z)$ .

where  $M_\tau, M_{\hat{\gamma}}, M_\psi \in \mathbb{M}_2(\mathbb{Z}/\ell^f\mathbb{Z})$ . Recall from Equation (1) that  $\sigma = \varphi \circ \tau \circ \gamma \circ \hat{\psi}/[\deg(\varphi)\deg(\tau)\deg(\psi)]$ . Therefore, the prover can compute  $\hat{\sigma} \circ \varphi \circ \tau(P_0)$  and  $\hat{\sigma} \circ \varphi \circ \tau(Q_0)$  by the following:

$$\begin{aligned}\hat{\sigma} \circ \varphi \circ \tau \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix} &= \frac{\psi \circ \hat{\gamma} \circ \hat{\tau} \circ \hat{\varphi} \circ \varphi \circ \tau}{[\deg(\varphi)\deg(\tau)\deg(\psi)]} \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix} \\ &= \frac{\psi \circ \hat{\gamma} \circ \hat{\tau} \circ \tau}{[\deg(\tau)\deg(\psi)]} \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix} \\ &= \frac{\psi \circ \hat{\gamma}}{[\deg(\psi)]} \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix}.\end{aligned}$$

Since  $\hat{\gamma} \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix} = M_{\hat{\gamma}} \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix}$ , one can deduce

$$\hat{\sigma} \circ \varphi \circ \tau \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix} = \frac{M_{\hat{\gamma}}}{[\deg(\psi)]} \cdot \psi \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix}.$$

It follows from  $\psi \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix} = M_\psi \begin{pmatrix} P_1 \\ Q_1 \end{pmatrix}$  that

$$\hat{\sigma} \circ \varphi \circ \tau \begin{pmatrix} P_0 \\ Q_0 \end{pmatrix} = \frac{M_{\hat{\gamma}} \cdot M_\psi}{[\deg(\psi)]} \begin{pmatrix} P_1 \\ Q_1 \end{pmatrix}.$$

Note that

$$M_\tau \cdot \begin{pmatrix} \hat{\sigma} \circ \varphi(\tau(P_0)) \\ \hat{\sigma} \circ \varphi(\tau(Q_0)) \end{pmatrix} = \hat{\sigma} \circ \varphi \left( M_\tau \cdot \begin{pmatrix} \tau(P_0) \\ \tau(Q_0) \end{pmatrix} \right) = \hat{\sigma} \circ \varphi \begin{pmatrix} P_A \\ Q_A \end{pmatrix}. \quad (4)$$

Therefore,

$$\hat{\sigma} \circ \varphi \begin{pmatrix} P_A \\ Q_A \end{pmatrix} = \frac{M_\tau \cdot M_{\hat{\gamma}} \cdot M_\psi}{[\deg(\psi)]} \begin{pmatrix} P_1 \\ Q_1 \end{pmatrix}.$$

Algorithm 5 summarizes the fast response using the above techniques. The signature is  $(E_1, q, M)$ , where  $q$  is the degree of  $\tau$  and  $M = \frac{M_\tau \cdot M_{\hat{\gamma}} \cdot M_\psi}{[\deg(\psi)]}$ . Since  $\varphi$  can be derived from  $E_1$  and the message, the verifier has access to  $E_2$ ,  $\varphi(P_A)$  and  $\varphi(Q_A)$ . The verifier accepts if  $(E_2, E_1, q, (\varphi(P_A), \varphi(Q_A)), (P_1^M, Q_1^M))$  correctly represents an isogeny from  $E_2$  to  $E_1$ , where  $(P_1^M, Q_1^M)^T = M \cdot (P_1, Q_1)^T$ . This is equivalent to prove that  $\sigma$  is an isogeny from  $E_1$  to  $E_2$ .

As shown above, the curve coefficient of  $E_2$  is not required for the response generation. Therefore, the prover does not need to construct or evaluate the challenge isogeny  $\varphi$ . Furthermore, the information related to the secret isogeny  $\tau$  (such as the action matrix  $M_\tau$ ) can be computed during key generation. As a result, the prover only needs to compute the commitment isogeny in the signing phase. All the other computations, such as the generation of the action matrix  $M_{\hat{\gamma}}$ , are executed over quaternions and linear algebra.

*Remark 4.* It should be noted that the signature size can be further compressed. For example, the verifier can recover the entire matrix  $M$  with only three entries

---

**Algorithm 5** FasterRespond

---

**Require:** The isogeny  $\tau' : E_0 \rightarrow E_A$  of degree  $\ell'^\bullet$ , the ideals  $I_\tau$  and  $I_{\tau'}$  associated to  $\tau$  and  $\tau'$  respectively, the ideal  $I_\psi$  associated to  $\psi$ , the isogeny  $\varphi : E_A \rightarrow E_2$  of degree  $\ell^f$ , and the action matrices  $M_\tau$  and  $M_\psi$  defined in Equation (3).

**Ensure:** The matrix  $M$  such that  $(\hat{\sigma} \circ \varphi(P_A), \hat{\sigma} \circ \varphi(Q_A))^T = M \cdot (P_1, Q_1)^T$  and the degree  $q$  of the isogeny  $\sigma : E_1 \rightarrow E_2$ .

- 1:  $I_\varphi \leftarrow \mathbf{IsogenyToIdeal}(\ker(\varphi), \tau', I_{\tau'}), J \leftarrow \overline{I_\psi} \cdot I_\tau \cdot I_\varphi;$
  - 2:  $I \leftarrow \mathbf{RandomEquivalentIdeal}_{\ell_1^g}(J)$  and compute the reduced norm  $q$  of  $I$ ;
  - 3: If  $q$  is not  $\ell^g$ -good or  $\gcd(q, \ell') \neq 1$ , go back to Line 2;
  - 4: Compute  $\gamma \in \mathcal{O}$  such that  $\mathcal{O}\gamma = I_\psi \cdot I \cdot \overline{I_\tau \cdot I_\varphi}$ ;
  - 5: Compute the action matrix  $M_\gamma$  as defined in Equation (3);
  - 6:  $M \leftarrow \frac{M_\tau \cdot M_\gamma \cdot M_\psi}{[\deg(\psi)]};$
  - 7: **return**  $(M, q)$ .
- 

of the action matrix  $M$  according to the techniques in [11, Section 6.1]. Furthermore, to verify the validity of the representation the prover can only reveal the actions of the response isogeny on an  $\ell^{\lceil g/2 \rceil}$ -torsion basis. This halves the storage cost of  $M$ . In the meantime, one can set  $g \leq 2f$  instead of  $g \leq f$  when utilizing  $\mathbf{RandomEquivalentIdeal}_{\ell^g}$  to generate  $I_\sigma$ .<sup>6</sup>

---

<sup>6</sup> For efficiency, it is best to set  $2f \geq g + 4$ . See [11, Section 4.3, Section 4.4] for more details.